Scientific computing for geophysical problems

Observational seismology

Léonard Seydoux and Alexandre Fournier (seydoux,fournier@ipgp.fr)
October 2026 at the institut de physique du globe de Paris.



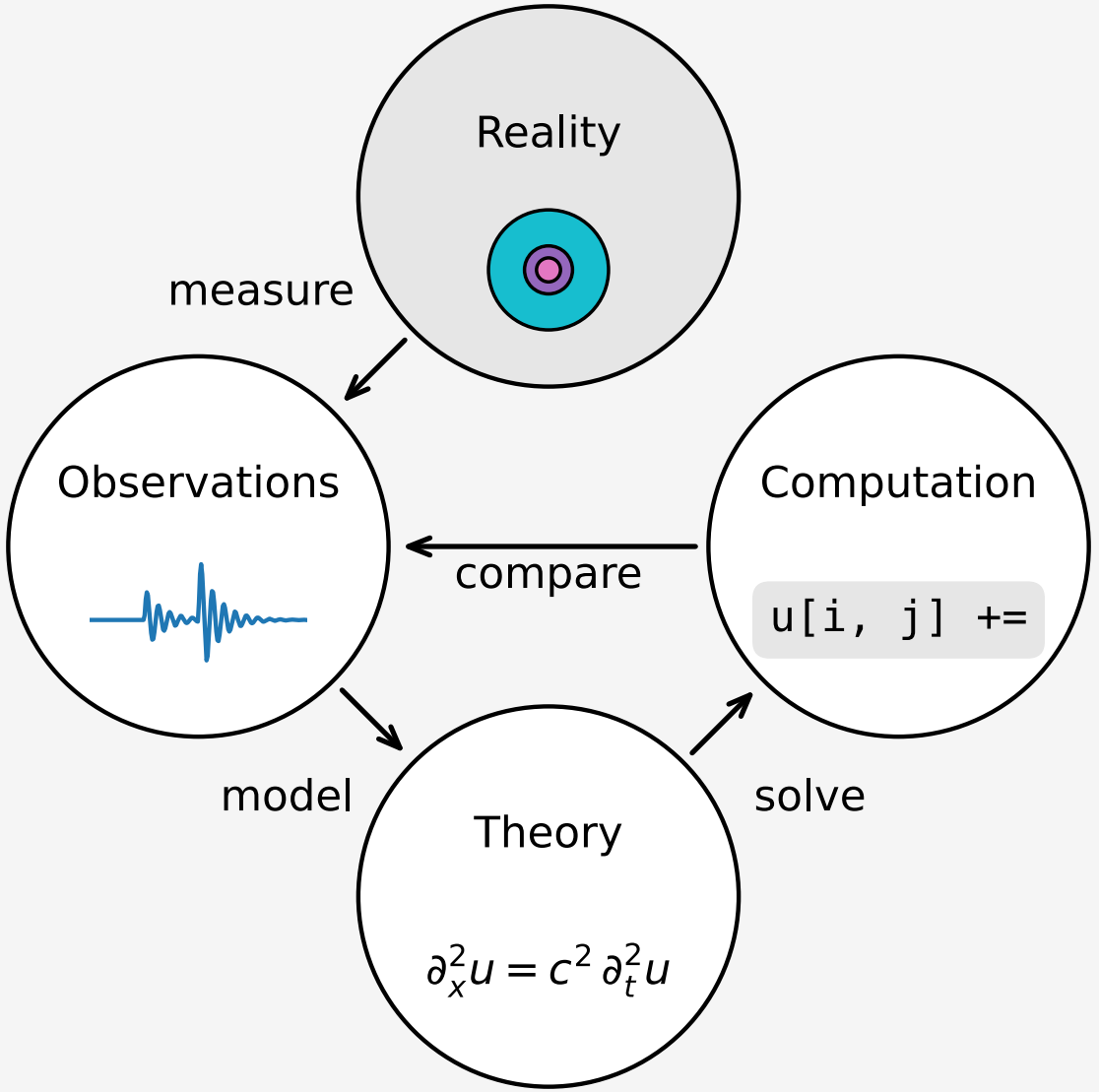




Learning outcomes

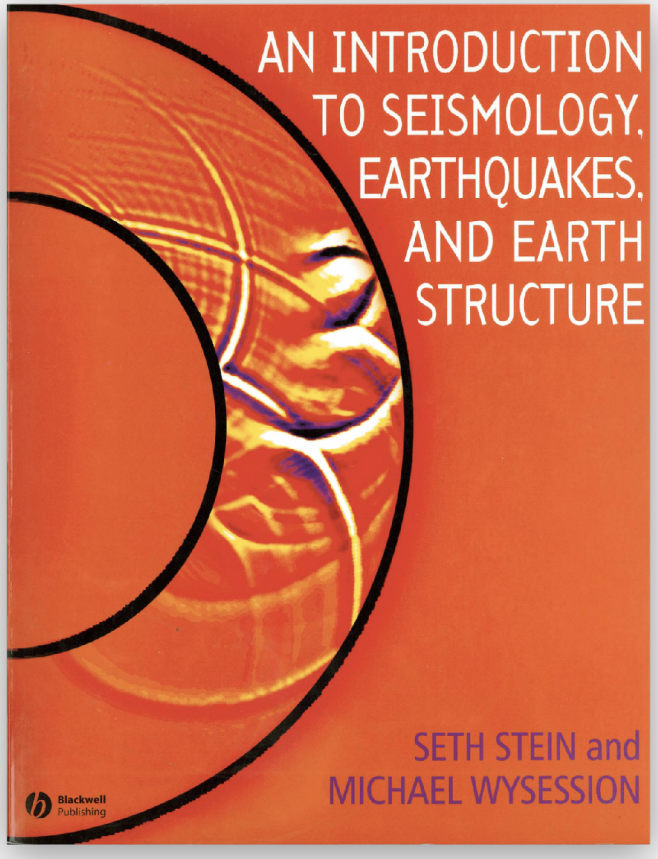
This session will teach you:

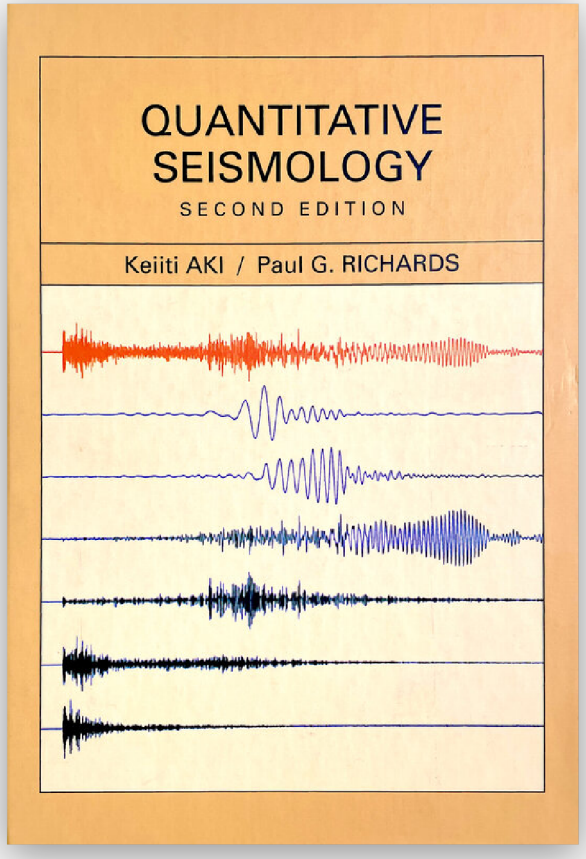
- Where seismic data come from, and how to get them
- How to turn raw counts into ground motion
- How to look at a seismogram in frequency: spectrum, filters, spectrogram
- The uncertainty principle between time and frequency
- How to recognize seismic waves in a seismogram and a spectrogram



Reference textbooks

- Stein & Wysession (2003): the reference for this class, and most figures (free to use for educational purposes)
- Aki & Richards (2002): when you need the full derivations

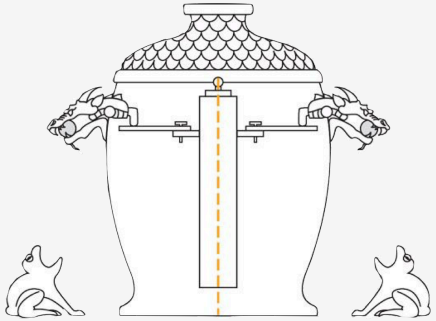


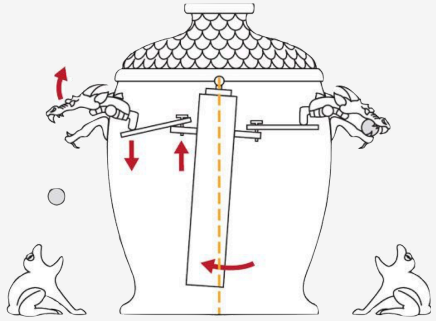


Stein, S., & Wysession, M. (2003). *An introduction to seismology, earthquakes, and Earth structure*. Blackwell Publishing. Aki, K., & Richards, P. G. (2002). *Quantitative seismology* (2nd ed.). University Science Books.

The first seismoscope

Built in China in 132 by Zhang Heng, it may have detected an unfelt earthquake in 143

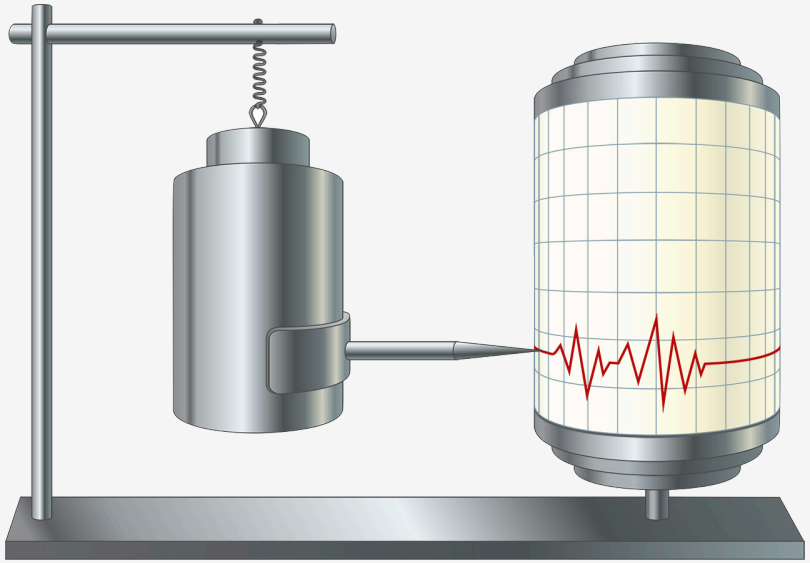




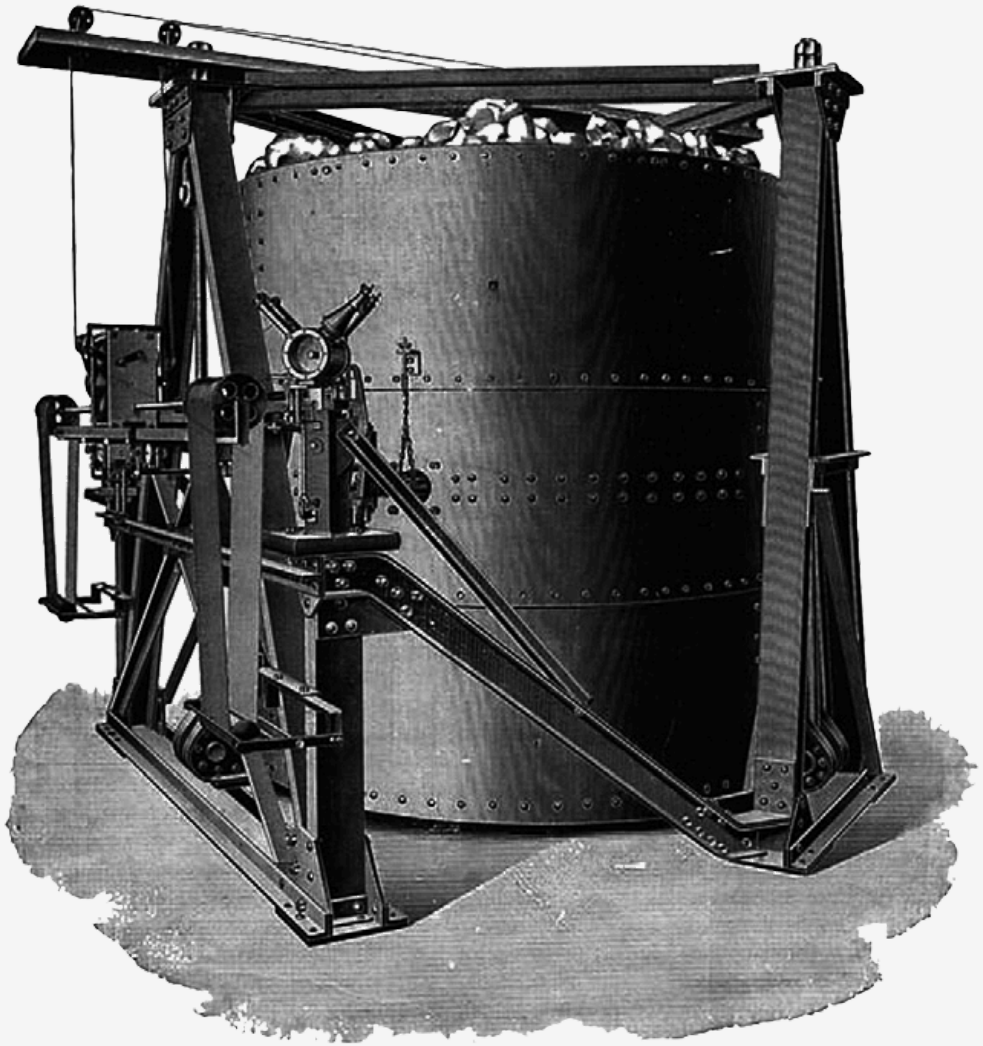


Early seismometers

A pendulum stays still while the ground shakes, and records motion.



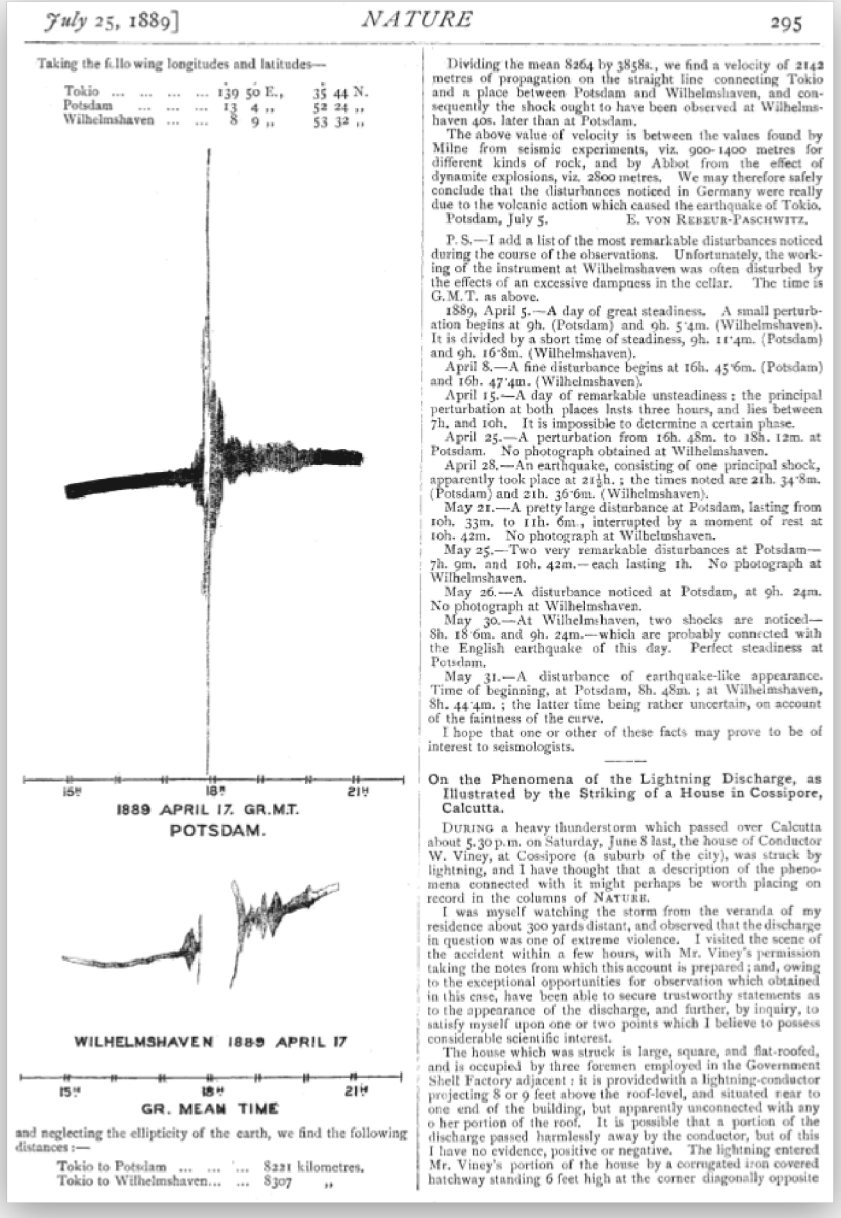
Example: Wiechert seismometer (1904), with a 17-ton pendulum



Scheme from seis-insight.eu, sketch from Ritter et al. (2000)

First record of a distant earthquake

An earthquake in **Japan** recorded in **Potsdam**, Germany, in 1889 by von Rebeur-Paschwitz (9,000 km away).



July 25, 1889] NATURE 295

Taking the following longitudes and latitudes—

Tokio	139° 50′ E.,	35° 44′ N.
Potsdam	13° 4′ ,,	52° 24′ ,,
Wilhelmshaven	8° 9′ ,,	53° 32′ ,,

1889 APRIL 17. GR.M.T.
POTSDAM.

WILHELMSHAVEN 1889 APRIL 17

GR. MEAN TIME

and neglecting the ellipticity of the earth, we find the following distances :—

Tokio to Potsdam	8221 kilometres.
Tokio to Wilhelmshaven	8307 ,,

Dividing the mean 8264 by 3858s., we find a velocity of 2142 metres of propagation on the straight line connecting Tokio and a place between Potsdam and Wilhelmshaven, and consequently the shock ought to have been observed at Wilhelmshaven 40s. later than at Potsdam.

The above value of velocity is between the values found by Milne from seismic experiments, viz. 900-1400 metres for different kinds of rock, and by Abbot from the effect of dynamite explosions, viz. 2800 metres. We may therefore safely conclude that the disturbances noticed in Germany were really due to the volcanic action which caused the earthquake of Tokio.

Potsdam, July 5. E. VON REBEUR-PASCHWITZ.

P. S.—I add a list of the most remarkable disturbances noticed during the course of the observations. Unfortunately, the working of the instrument at Wilhelmshaven was often disturbed by the effects of an excessive dampness in the cellar. The time is G.M.T. as above.

1889, April 5.—A day of great steadiness. A small perturbation begins at 9h. (Potsdam) and 9h. 5·4m. (Wilhelmshaven). It is divided by a short time of steadiness, 9h. 11·4m. (Potsdam) and 9h. 16·8m. (Wilhelmshaven).

April 8.—A fine disturbance begins at 16h. 45·6m. (Potsdam) and 16h. 47·4m. (Wilhelmshaven).

April 15.—A day of remarkable unsteadiness : the principal perturbation at both places lasts three hours, and lies between 7h. and 10h. It is impossible to determine a certain phase.

April 25.—A perturbation from 16h. 48m. to 18h. 12m. at Potsdam. No photograph obtained at Wilhelmshaven.

April 28.—An earthquake, consisting of one principal shock, apparently took place at 21½h. ; the times noted are 21h. 34·8m. (Potsdam) and 21h. 36·6m. (Wilhelmshaven).

May 21.—A pretty large disturbance at Potsdam, lasting from 10h. 33m. to 11h. 6m., interrupted by a moment of rest at 10h. 42m. No photograph at Wilhelmshaven.

May 25.—Two very remarkable disturbances at Potsdam—7h. 9m. and 10h. 42m.—each lasting 1h. No photograph at Wilhelmshaven.

May 26.—A disturbance noticed at Potsdam, at 9h. 24m. No photograph at Wilhelmshaven.

May 30.—At Wilhelmshaven, two shocks are noticed—8h. 18·6m. and 9h. 24m.—which are probably connected with the English earthquake of this day. Perfect steadiness at Potsdam.

May 31.—A disturbance of earthquake-like appearance. Time of beginning, at Potsdam, 8h. 48m. ; at Wilhelmshaven, 8h. 44·4m. ; the latter time being rather uncertain, on account of the faintness of the curve.

I hope that one or other of these facts may prove to be of interest to seismologists.

On the Phenomena of the Lightning Discharge, as Illustrated by the Striking of a House in Cossipore, Calcutta.

DURING a heavy thunderstorm which passed over Calcutta about 5.30 p.m. on Saturday, June 8 last, the house of Conductor W. Viney, at Cossipore (a suburb of the city), was struck by lightning, and I have thought that a description of the phenomena connected with it might perhaps be worth placing on record in the columns of NATURE.

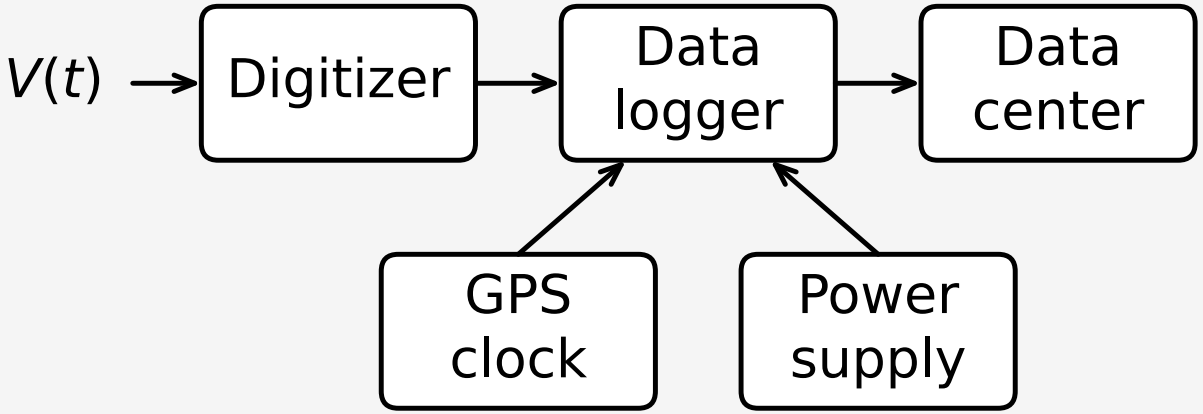
I was myself watching the storm from the veranda of my residence about 300 yards distant, and observed that the discharge in question was one of extreme violence. I visited the scene of the accident within a few hours, with Mr. Viney's permission taking the notes from which this account is prepared ; and, owing to the exceptional opportunities for observation which obtained in this case, have been able to secure trustworthy statements as to the appearance of the discharge, and further, by inquiry, to satisfy myself upon one or two points which I believe to possess considerable scientific interest.

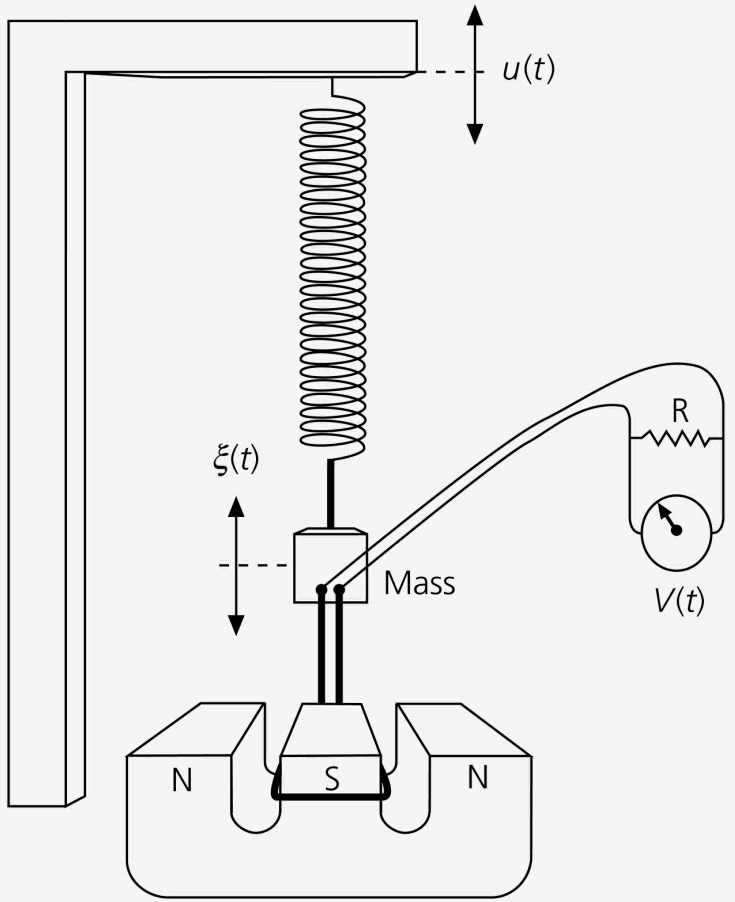
The house which was struck is large, square, and flat-roofed, and is occupied by three foremen employed in the Government Shell Factory adjacent : it is provided with a lightning-conductor projecting 8 or 9 feet above the roof-level, and situated near to one end of the building, but apparently unconnected with any other portion of the roof. It is possible that a portion of the discharge passed harmlessly away by the conductor, but of this I have no evidence, positive or negative. The lightning entered Mr. Viney's portion of the house by a corrugated iron covered hatchway standing 6 feet high at the corner diagonally opposite

From Nature, July 25, 1889

Modern seismometers

A feedback force keeps the mass in place: the required voltage $V(t)$ is the recorded signal





$V(t)$ is then digitized and sent to datacenters

Stein & Wysession (2003), fig. 6.6-5

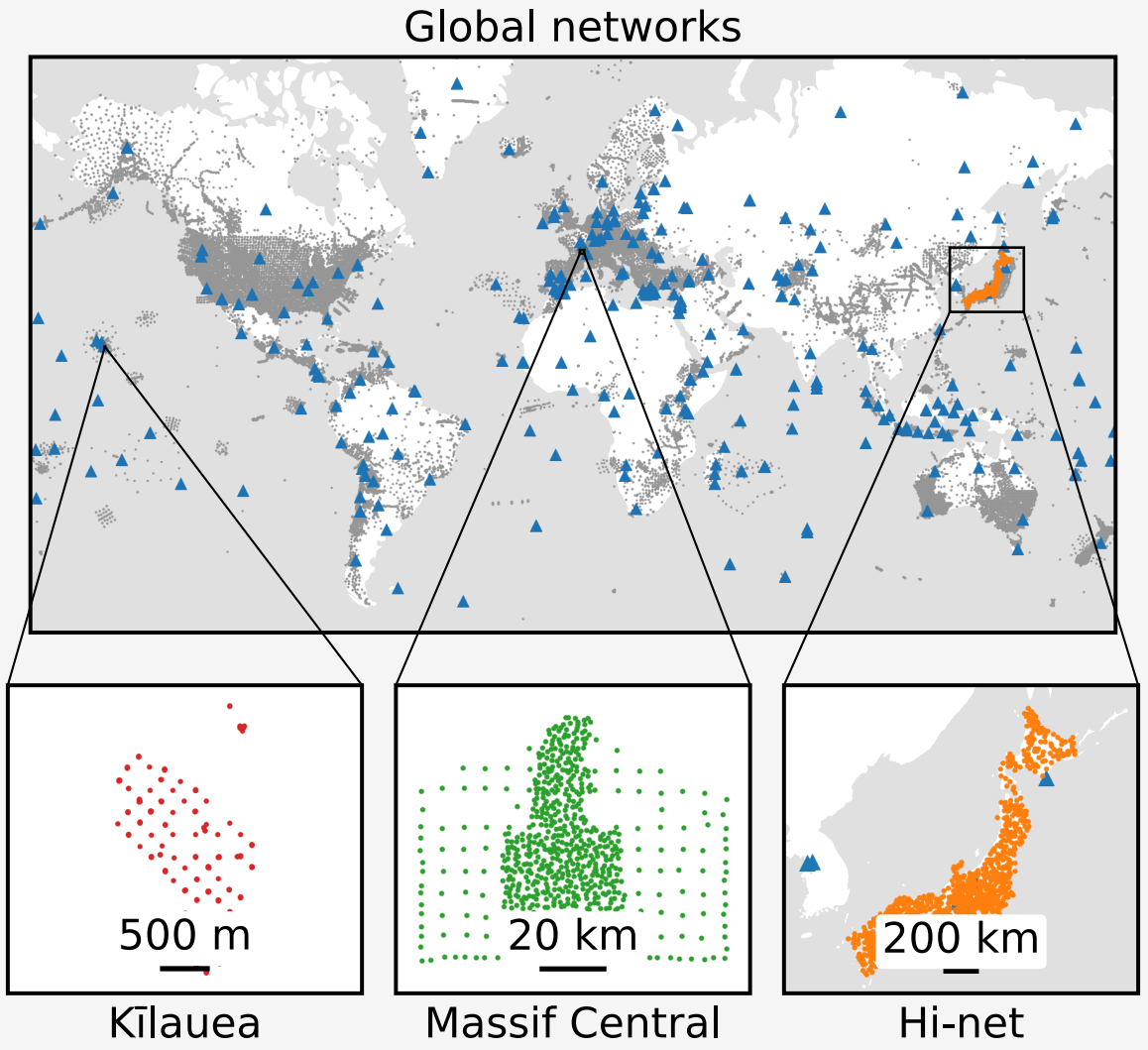
Modern seismic station



Station site in Northern California, © David Croker, USGS, vault and sensor (left), solar panels (right)

Modern seismic networks

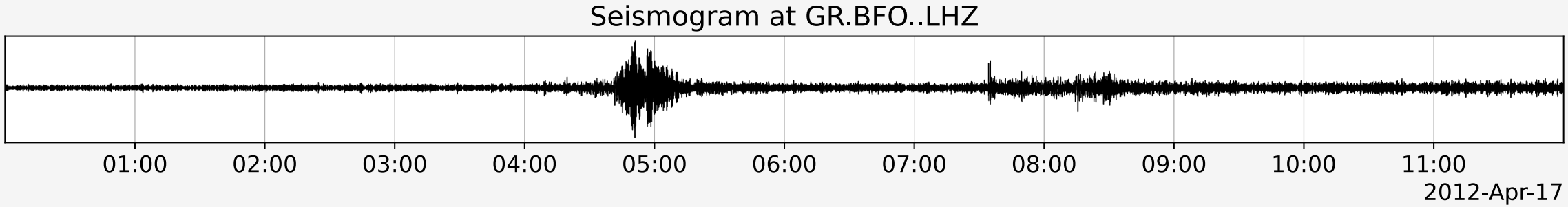
- 10,000+ permanent
- Many more temporary
- From meters to the whole globe
- Large volumes of data
- Mostly open access



All stations with open metadata at the FDSN data centers (EarthScope federated catalog, GeoNet, NRCan, BGS, KAGSR, RSBR, GEOSHAKE) and of the JMA bulletin (Japan). Global networks: GSN (II, IU), GEOSCOPE (G) and GEOFON (GE)

What is a seismogram?

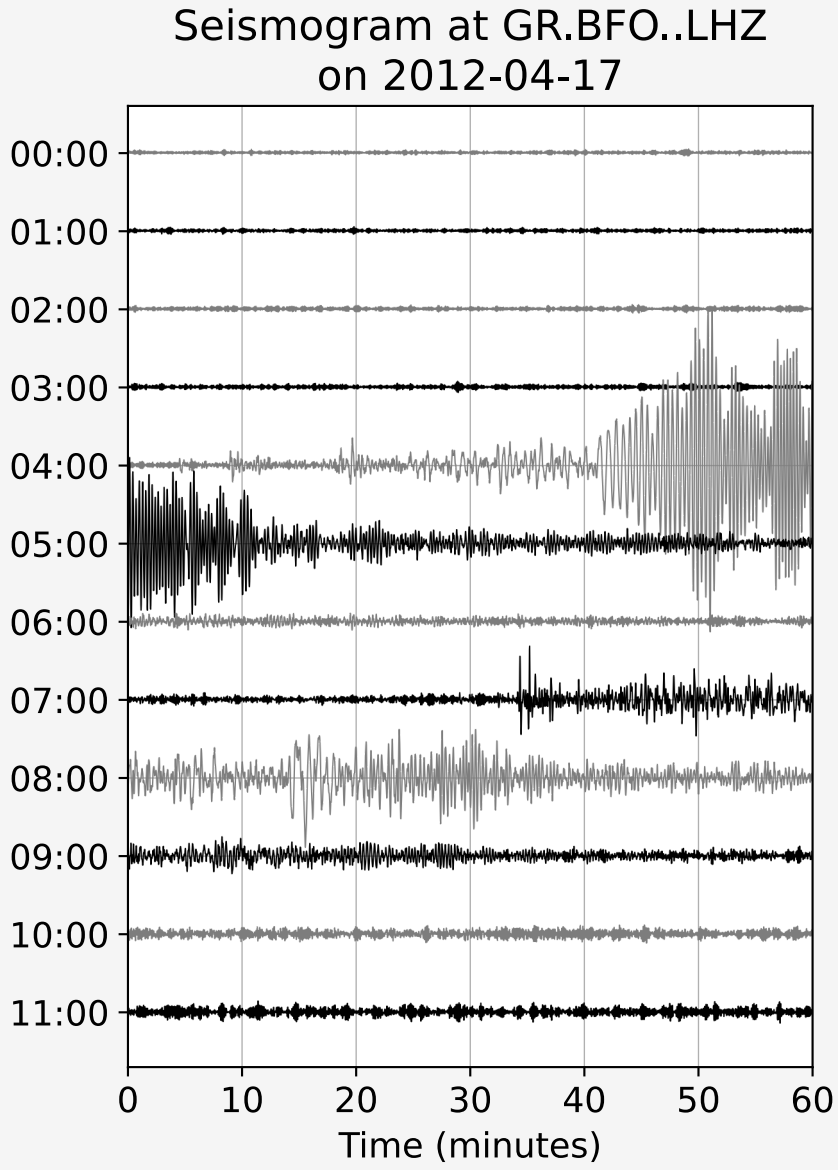
Seismology (from Ancient Greek σεισμός, "earthquake", and -λογία, "study of") is the scientific study of earthquakes and the propagation of elastic waves through the Earth or through other planet-like bodies. It also includes studies of earthquake environmental effects such as tsunamis, as well as diverse seismic sources such as volcanic, tectonic, glacial, fluvial, oceanic, atmospheric, and artificial processes such as explosions. A related field that uses geology to infer information regarding past earthquakes is paleoseismology. A recording of Earth motion as a function of time is called a **seismogram**.



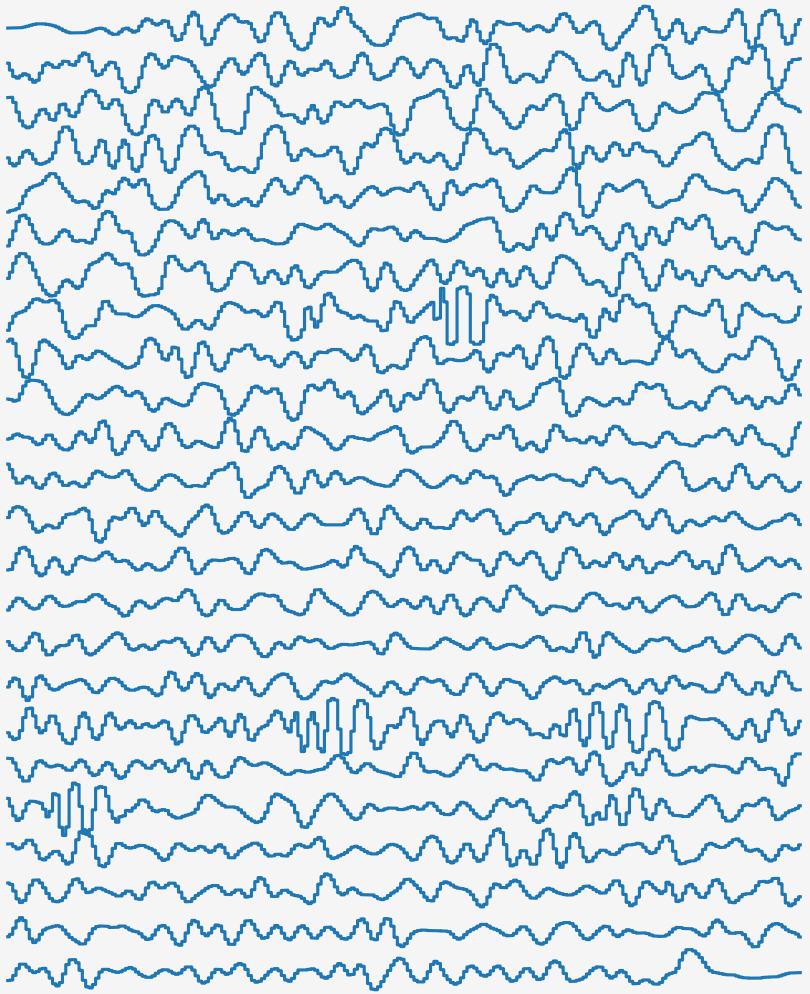
Twelve hours of ground motion

A twelve-hour seismogram at station BFO, Germany: earthquakes, noise, and much more

Most of what we know about the Earth's interior comes from such records



Digital seismic data

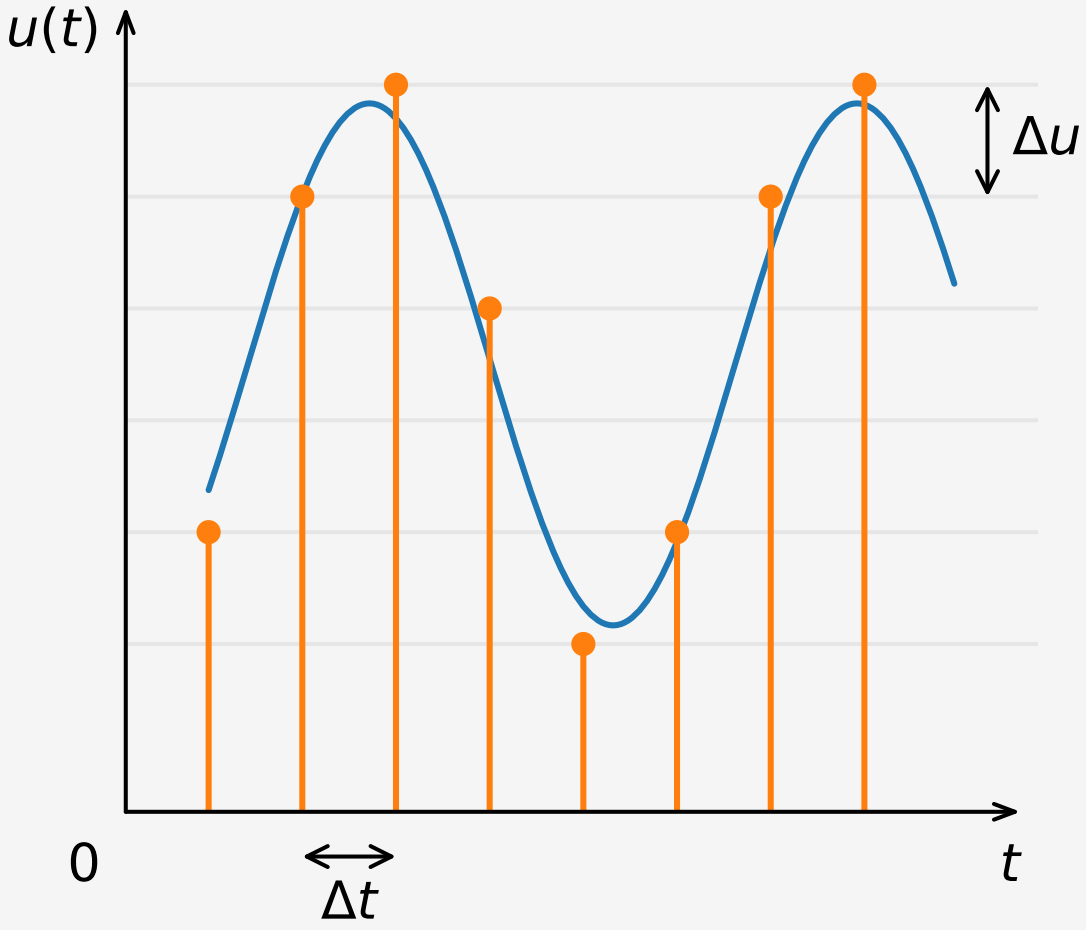


Digital seismic data

The ground motion $u(t)$ is sampled every Δt

$$u[n] = u(n\Delta t), \quad n = 0 \dots N-1$$

- Number of samples: N
- Sampling rate: $f_s = 1/\Delta t$, in Hz (or samples per second)
- Duration: $T = N\Delta t = N/f_s$



How much data?

Record	Samples per day	Volume per day
One channel at 100 Hz	8.6 million	35 MB
One three-component station	26 million	100 MB
One station over a year	9.5 billion	38 GB
One fiber, 10,000 channels at 1 kHz (DAS)	860 billion	3.5 TB

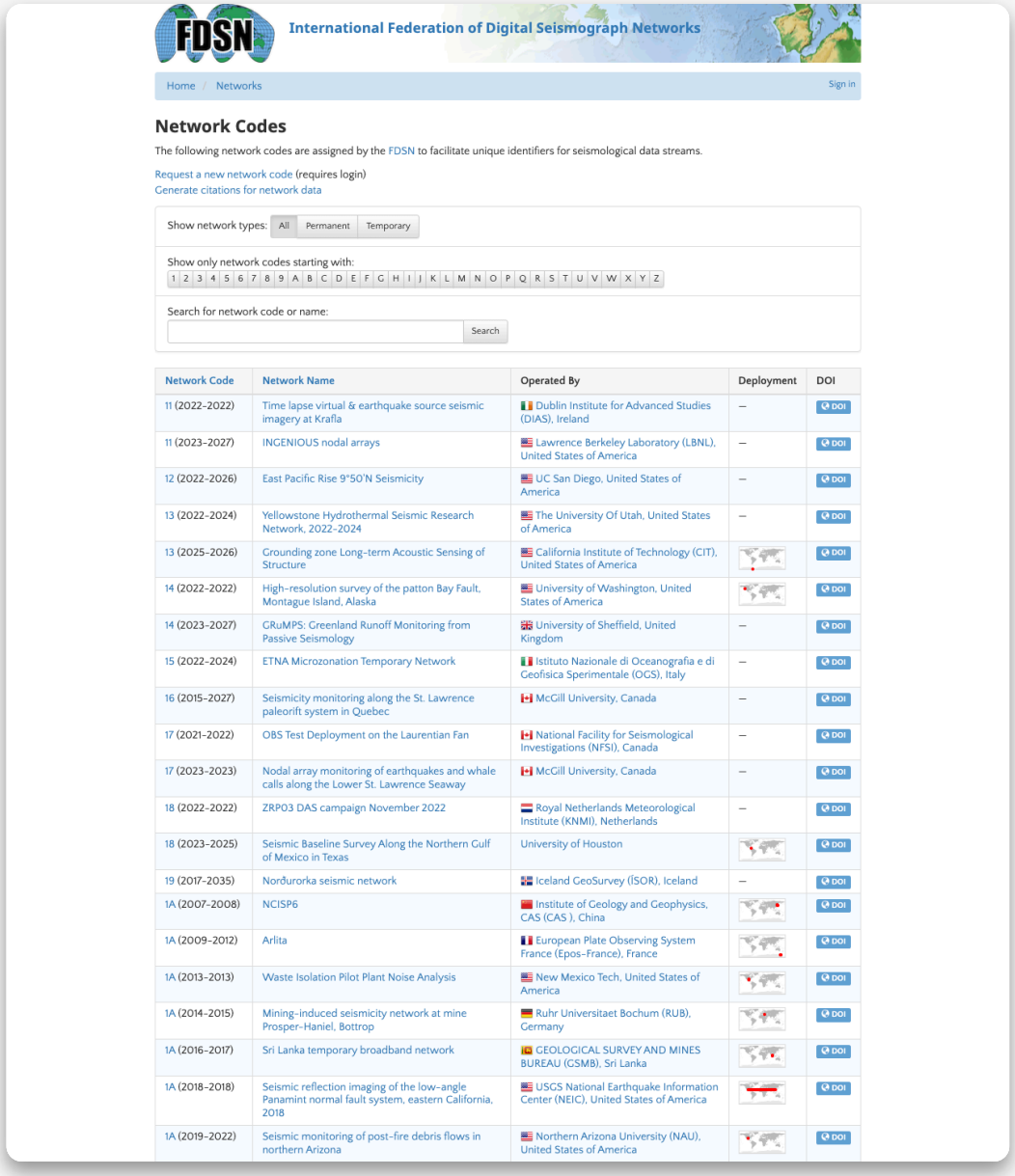
Volumes for 4-byte samples, before compression

Data centers

The International Federation of Digital Seismograph Networks (FDSN)

- Every network has a unique code and digital object identifier (DOI)
- Map, coordinates, sensors and operating range of all networks

The EarthScope Consortium (formerly IRIS) is one of the largest data centers. IPGP, RESIF, SCEDC, NCEDC or GFZ are also accessible with ObsPy



Seismic data in Python

- `Stream` : a list of traces
- `Trace` : one channel, with
  - `data` : the samples, as a NumPy array
  - `stats` : identifier, sampling rate, start time
- `Inventory` : stations, channels and instrument responses

Formats: miniSEED (waveforms), StationXML (metadata), SAC (one trace per file)

```
1  from obspy import read
2
3  stream = read("record.mseed")
4  trace = stream[0]
5  trace.stats.sampling_rate
6  trace.data
```

Naming seismic data

Every record has a unique identifier, as `IU.CTAO.00.BHZ`

Code	Part	Meaning
`IU`	Network	Global Seismograph Network
`CTAO`	Station	Charters Towers, Australia
`00`	Location	Sensor of the station (`00`, `10`…)
`B`	Band	Broadband, 10–80 Hz (`L`: 1 Hz, `H`: ≥ 80 Hz)
`H`	Instrument	Seismometer (`N`: accelerometer)
`Z`	Component	Vertical (`N`, `E`: horizontal)

SEED channel naming convention, from the FDSN

Requesting data

One FDSN request is a URL: network, station, location, channel and time window

```
service.earthscope.org
/fdsnws/dataselect/1/query
?net=G&sta=SSB&loc=00&cha=BHZ
&starttime=2011-03-11T05:46:24
&endtime=2011-03-11T06:26:24
```

The same request works for any FDSN data center

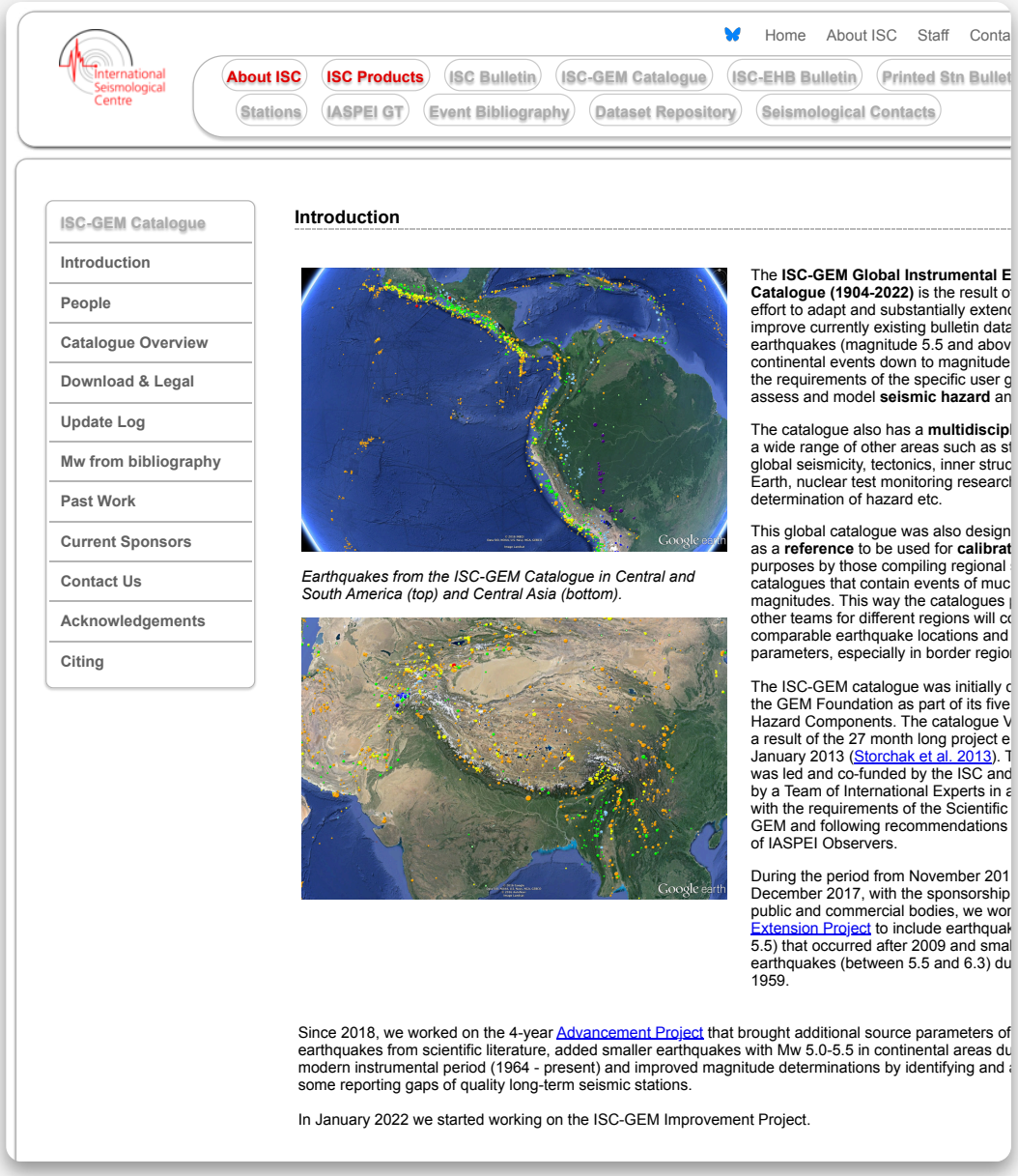
```python
from obspy import UTCDateTime
from obspy.clients.fdsn import Client

client = Client("EARTHSCOPE")
start = UTCDateTime("2011-03-11T05:46:24")
stream = client.get_waveforms(
    "G", "SSB", "00", "BHZ", start, start + 240
)
```

Finding events

Catalogs give the time, location and magnitude of events

- USGS ComCat: global
- ISC-GEM: large events since 1904
- EMSC: Euro-Mediterranean
- Regional and volcano observatories

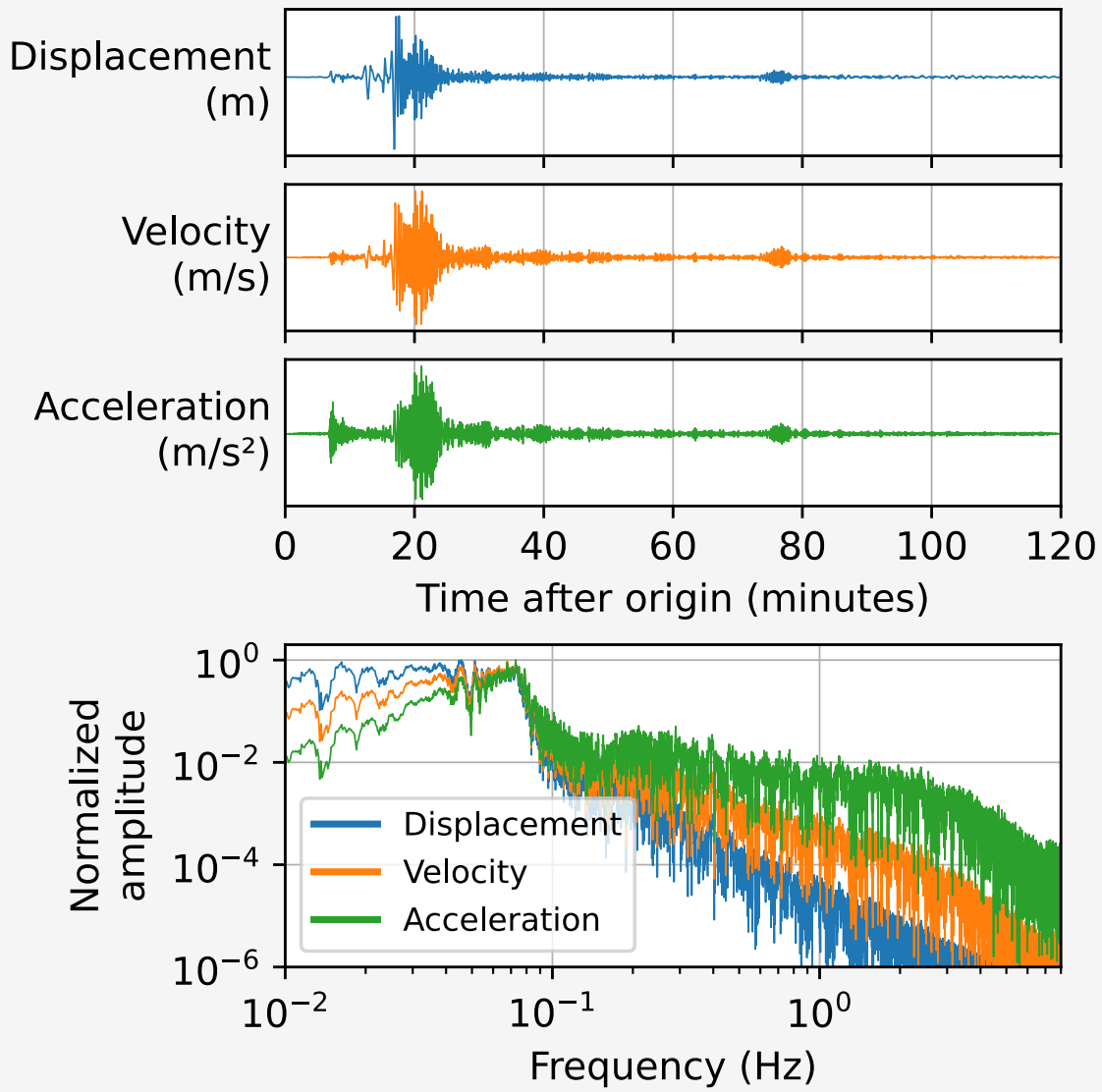


From counts to ground motion

Records are in counts: divide by the instrument response, in a band where it is large (`pre_filt`)
Each time derivative multiplies the spectrum by $2i\pi f$

$$\hat{v}(f) = 2i\pi f \hat{u}(f), \quad \hat{a}(f) = 2i\pi f \hat{v}(f)$$

- Displacement: low frequencies
- Acceleration: high frequencies

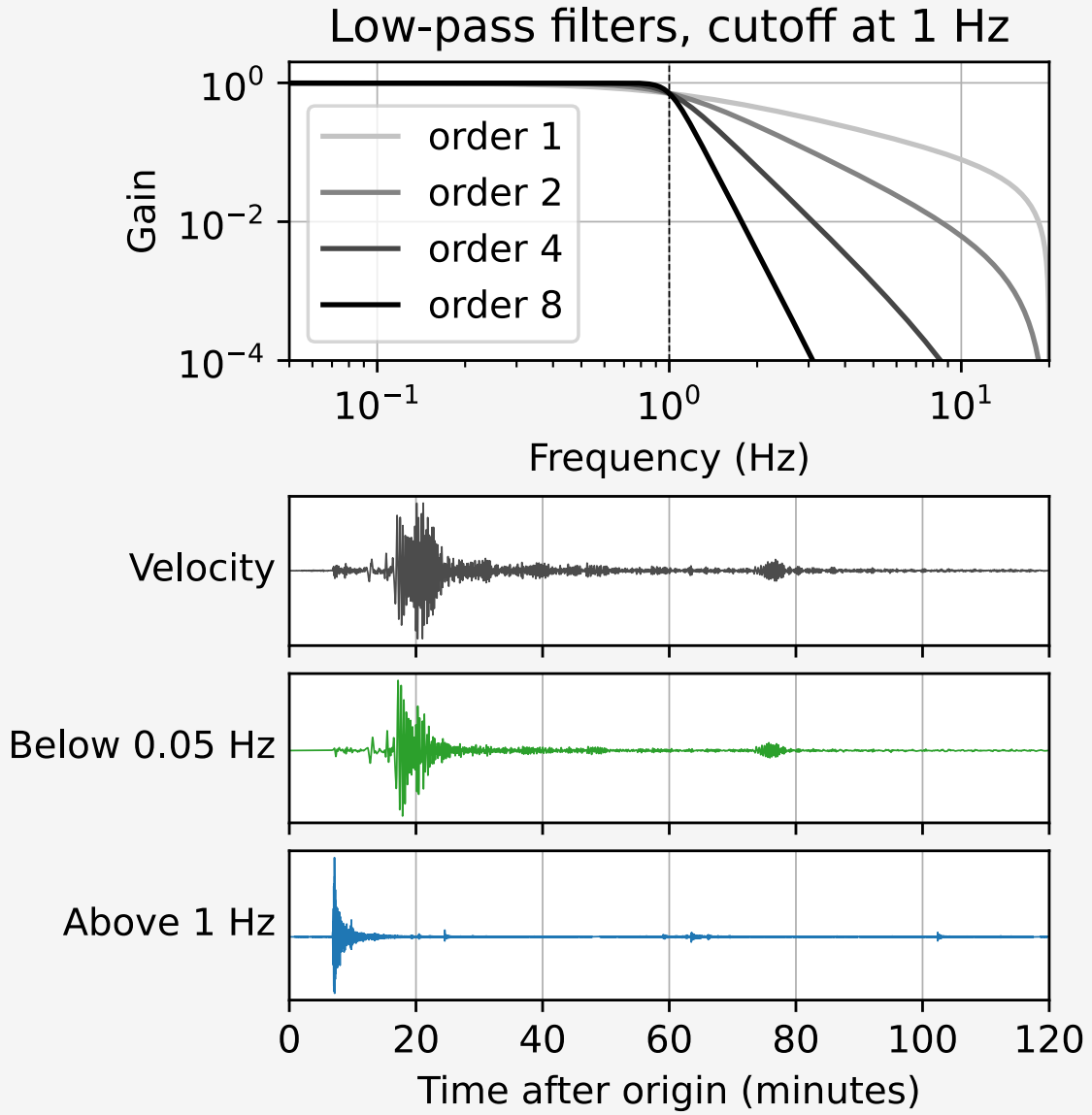


Kermadec Islands earthquake (Mw 8.1, 4 March 2021) recorded at IU.CTAO.00.BHZ, the example of the lab

Filtering

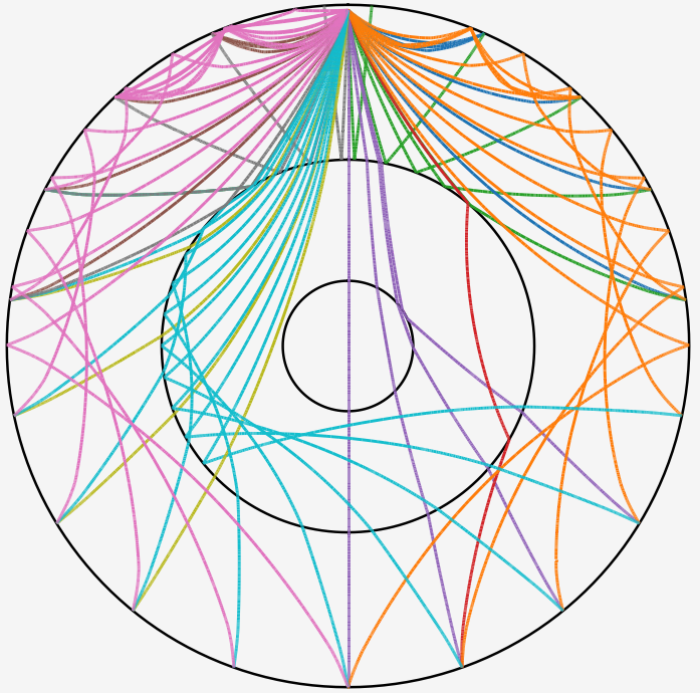
Keep a frequency band: low-pass, high-pass or band-pass

- Butterworth: flat in the band
- Higher order (`corners`): sharper cut
- `zerophase=True` : no phase shift
- Low frequencies: surface waves
- High frequencies: body waves



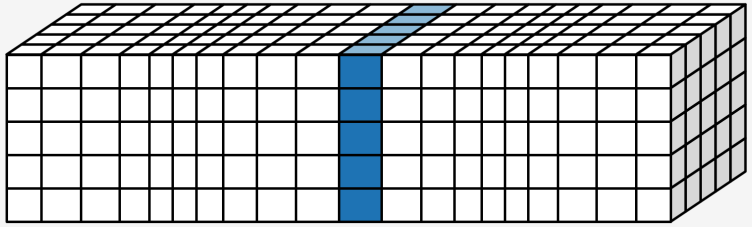
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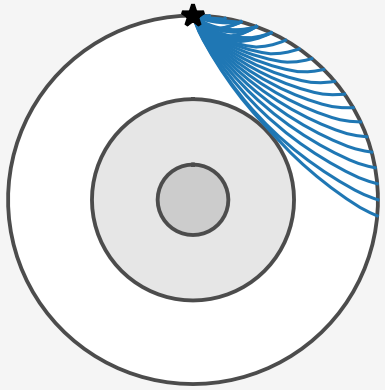
Seismic waves



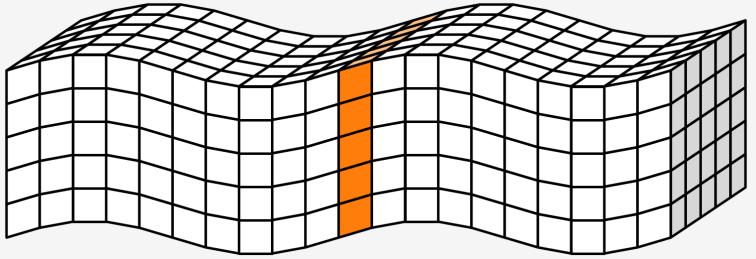
Body waves

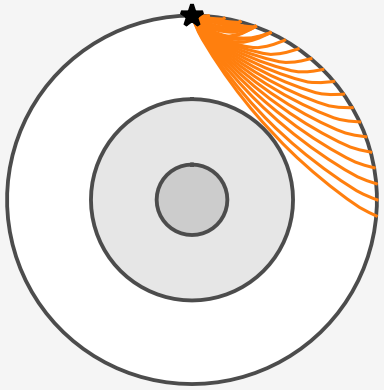
P wave: compressional wave





S wave: shear wave

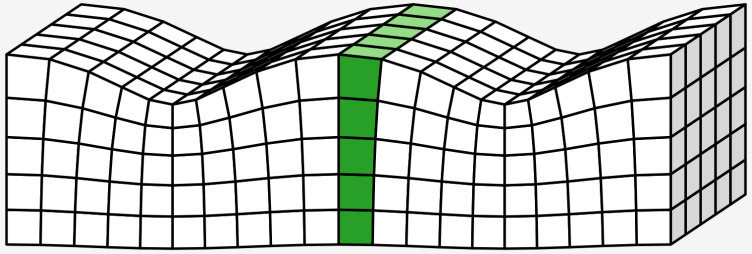




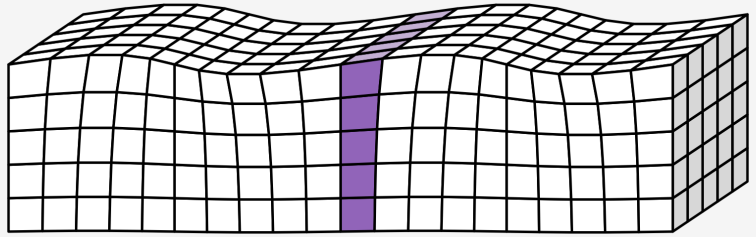
Ray paths in the iasp91 model, computed with ObsPy (TauP)

Surface waves

Rayleigh wave: rolling motion, decaying with depth



Love wave: horizontal shear, decaying with depth



Dispersion

Long periods reach deeper, faster rocks

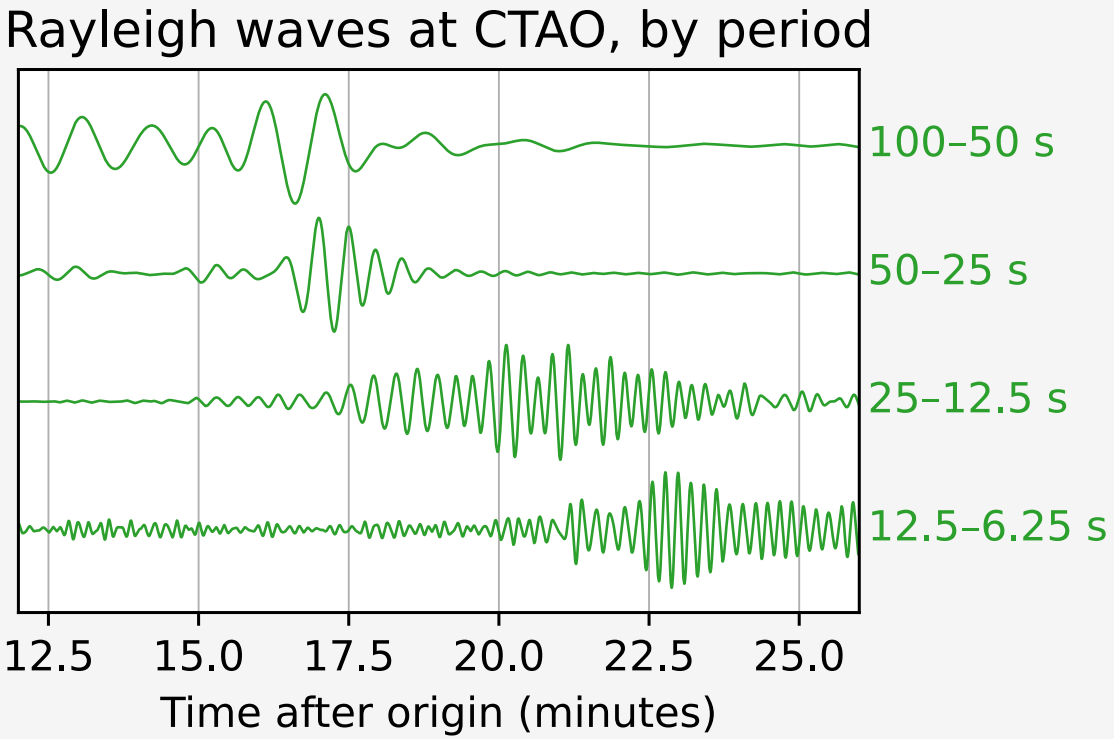
Phase and group velocities

$$v_\varphi = \frac{\omega}{k}, \quad v_g = \frac{d\omega}{dk}$$

Each period arrives at

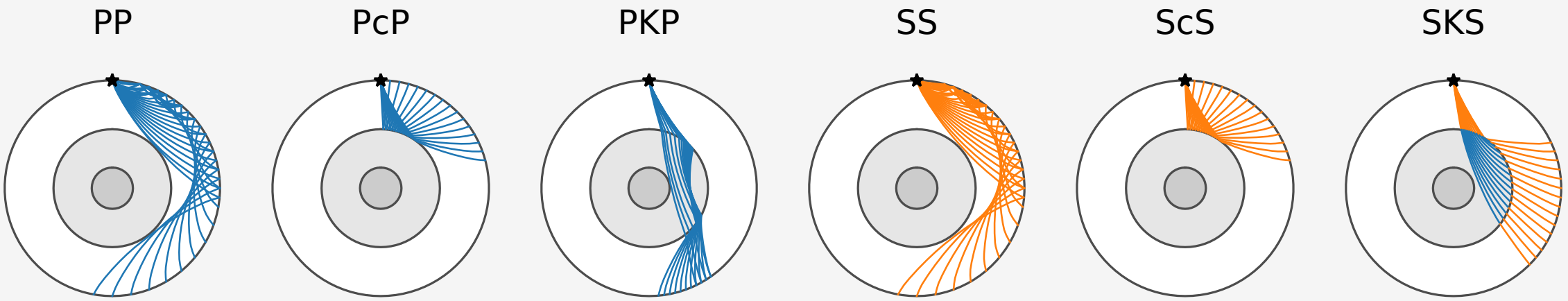
$$t(\omega) = \frac{\Delta}{v_g(\omega)}$$

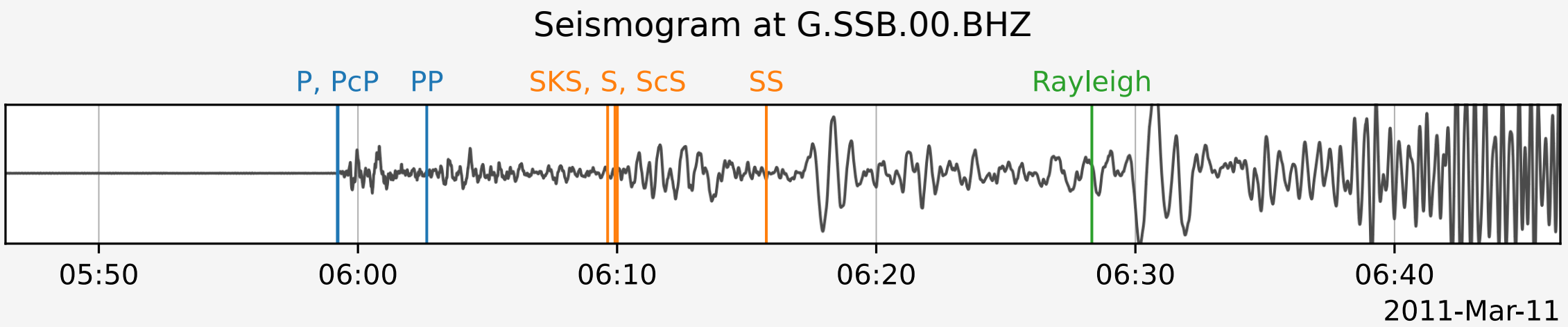
- Long periods first: in a spectrogram, the frequency increases with time
- v_g: a probe of the crust and mantle



Kermadec Islands earthquake (Mw 8.1, 4 March 2021) recorded at IU.CTAO.00.BHZ, the example of the lab, filtered in period bands

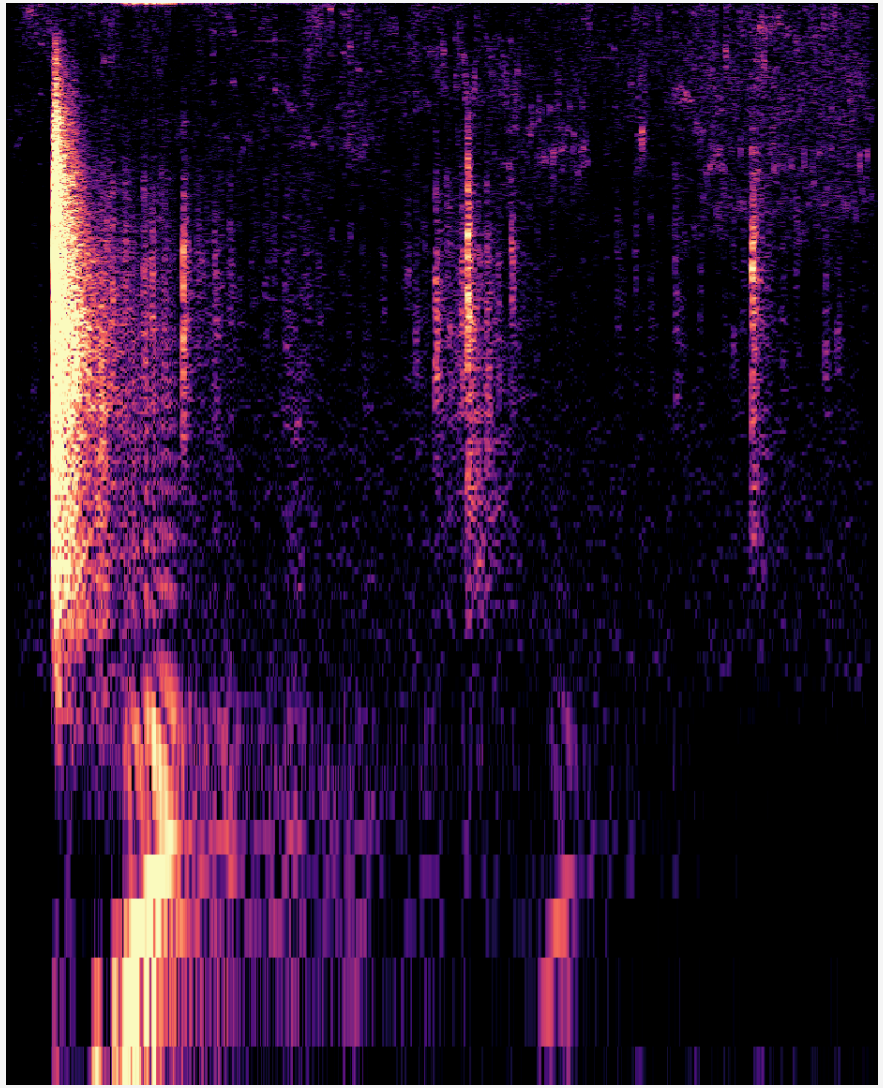
Seismic phases





Ray paths and arrival times in the iasp91 model, computed with ObsPy (TauP).
Tohoku-Oki earthquake (2011, Mw 9.1) recorded at station G.SSB, France

Time and frequency



Spectrum of a seismogram

The discrete Fourier transform of N samples, taken every Δt

$$\widehat{u}[k] = \sum_{n=0}^{N-1} u[n]\, e^{-2i\pi kn/N}$$

- Frequencies every $\Delta f = 1/T = 1/(N\Delta t)$
- Up to the Nyquist frequency $f_s/2$
- With NumPy: `numpy.fft.rfft` and `numpy.fft.rfftfreq`
- Shown as $|\widehat{u}(f)|$, in log-log scale

The spectrum tells which frequencies are in the record, not when

Building a spectrogram

Cut the record in windows of duration T, starting at times t_k

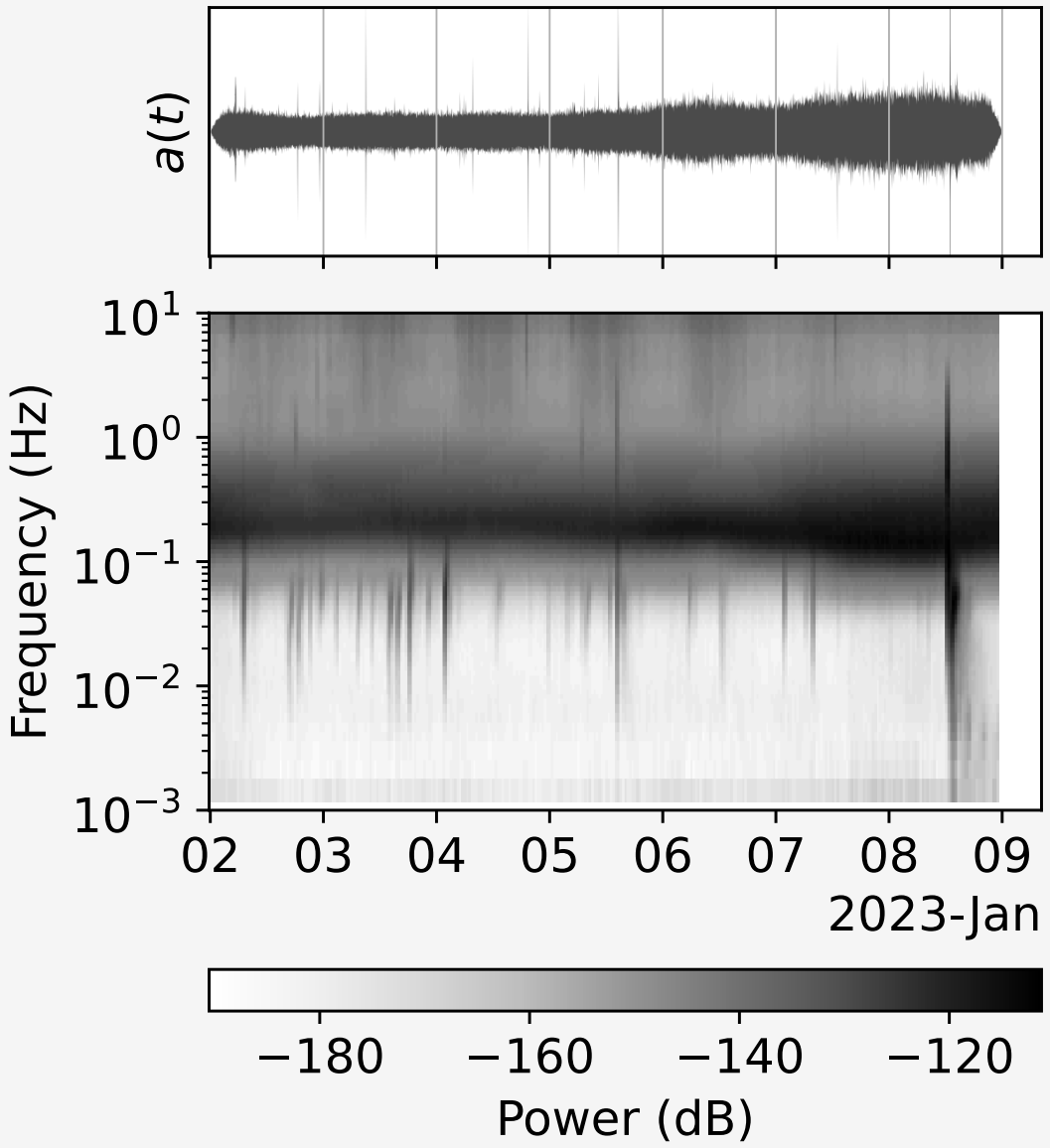
Fourier transform of each window, tapered by w

$$\hat{u}(t_k, f) = \int_0^T w(\tau)\, u(t_k + \tau)\, e^{-2i\pi f\tau} d\tau$$

Power spectral density, in decibels

$$P(t_k, f) = 10 \log_{10} \frac{2\,|\hat{u}(t_k, f)|^2}{T}$$

With `scipy.signal.spectrogram`:
`nperseg` samples per window, and
`noverlap` samples shared by two windows



Station G.SSB.00.BHZ, 2–8 January 2023, after removing the instrument response, with windows of 1 hour

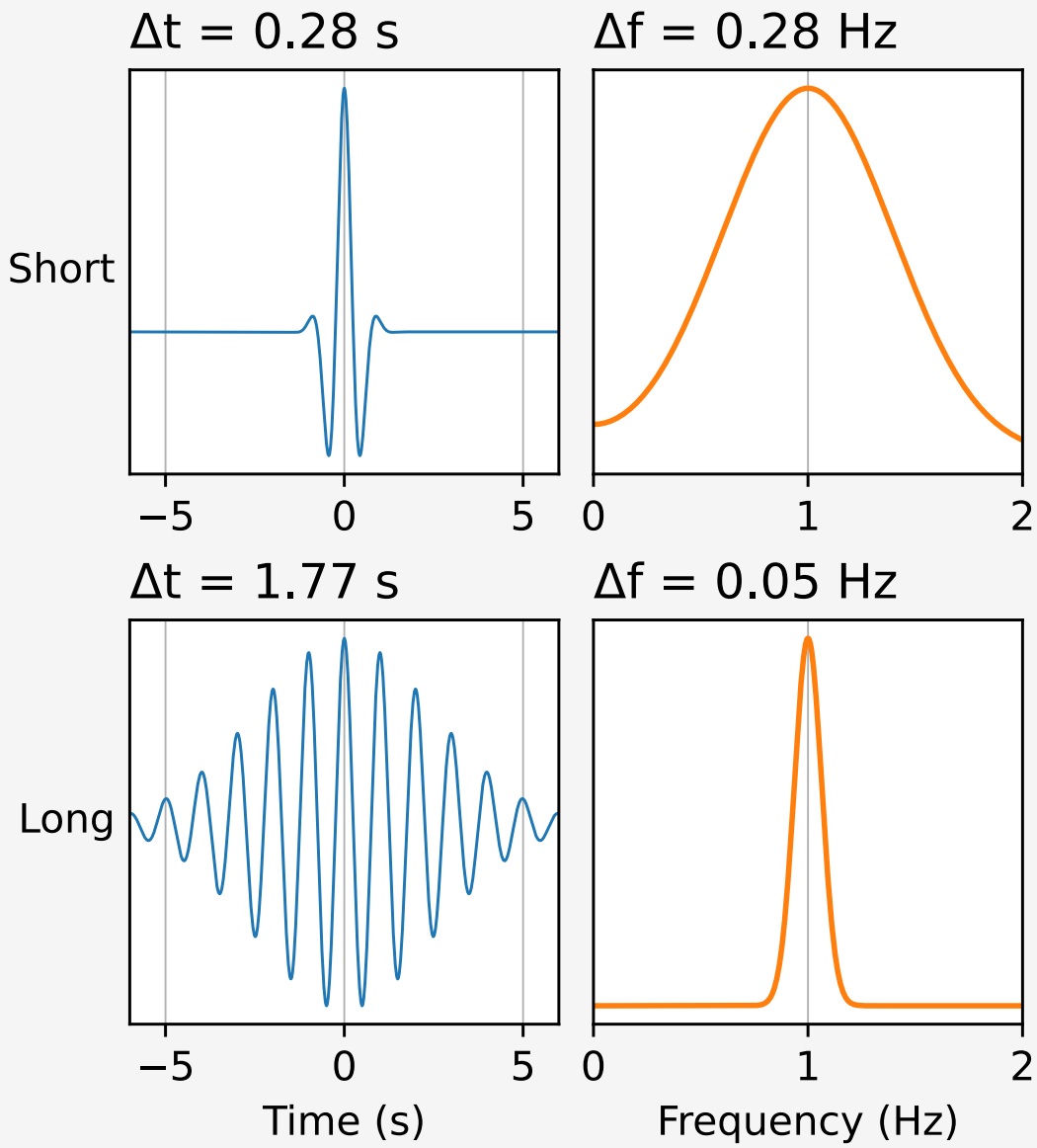
Uncertainty principle

A signal cannot be both short in time and narrow in frequency

$$\Delta t \, \Delta f \geq \frac{1}{4\pi}$$

with Δt and Δf the spreads (standard deviations) of $|u(t)|^2$ and $|\hat{u}(f)|^2$

- Short pulse: wide spectrum
- Long wave train: narrow spectrum
- The bound is reached for Gaussian envelopes



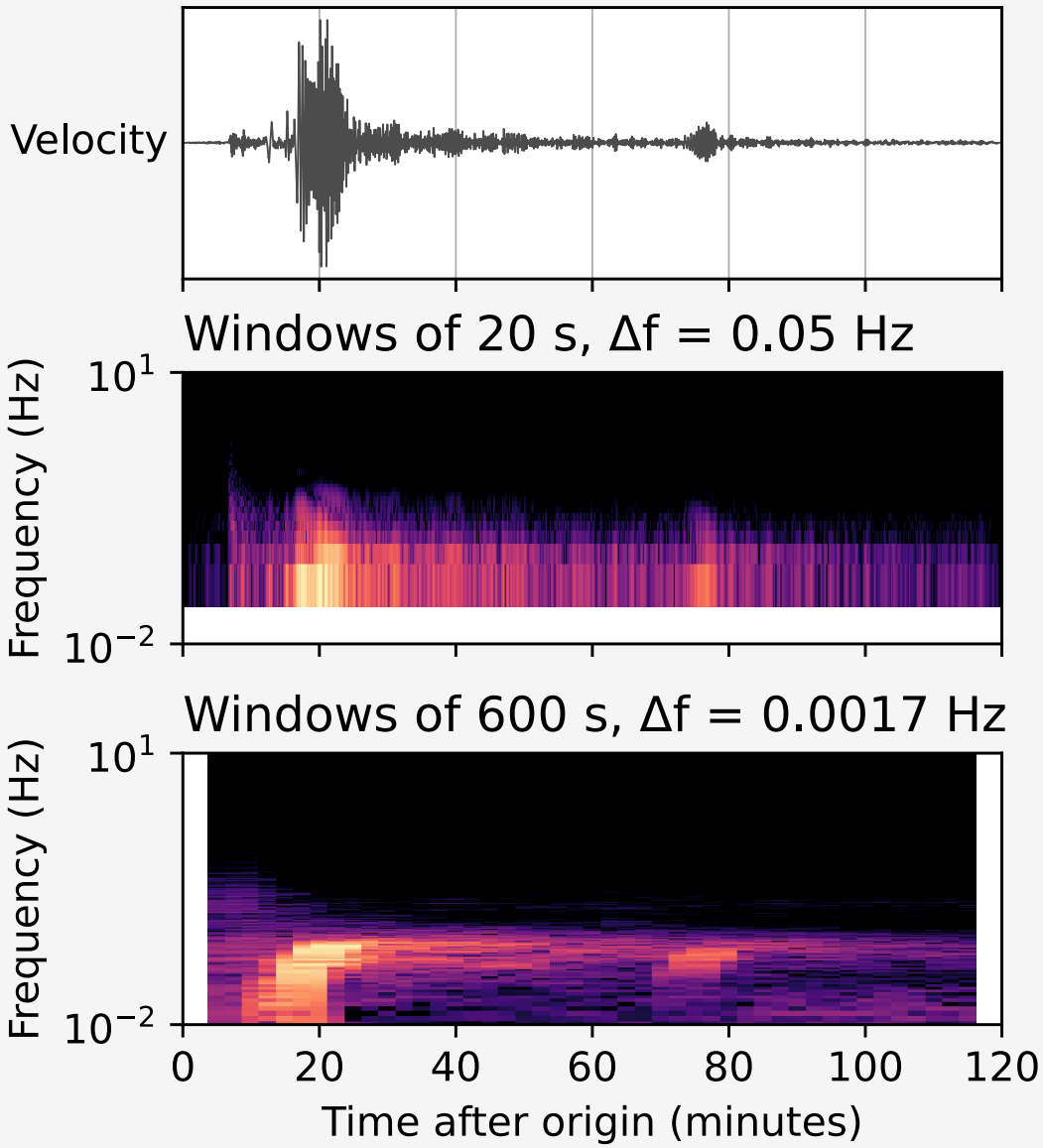
Gabor (1946), the same mathematics as Heisenberg's principle in quantum mechanics

Time–frequency trade-off

A window of duration T resolves $\Delta f = 1/T$ in frequency, and T in time

- Short windows: sharp in time, blurred in frequency
- Long windows: sharp in frequency, blurred in time
- Overlap (`noverlap`): smoother, not sharper

Choose T for what you want to see: arrival times or frequencies

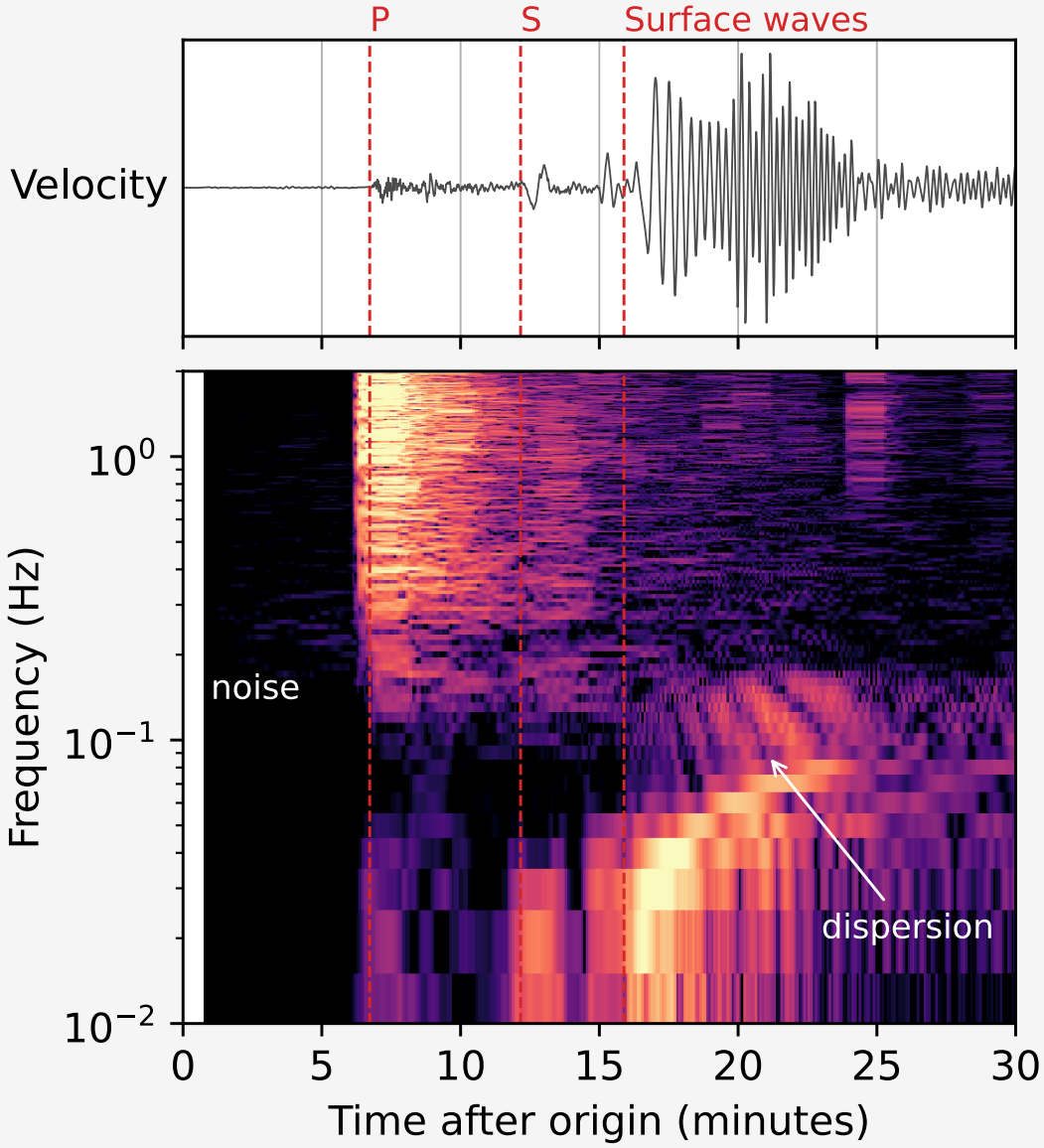


Kermadec Islands earthquake (Mw 8.1, 4 March 2021) recorded at IU.CTAO.00.BHZ, the example of the lab

Reading a spectrogram

The event of the lab, recorded 3,800 km away

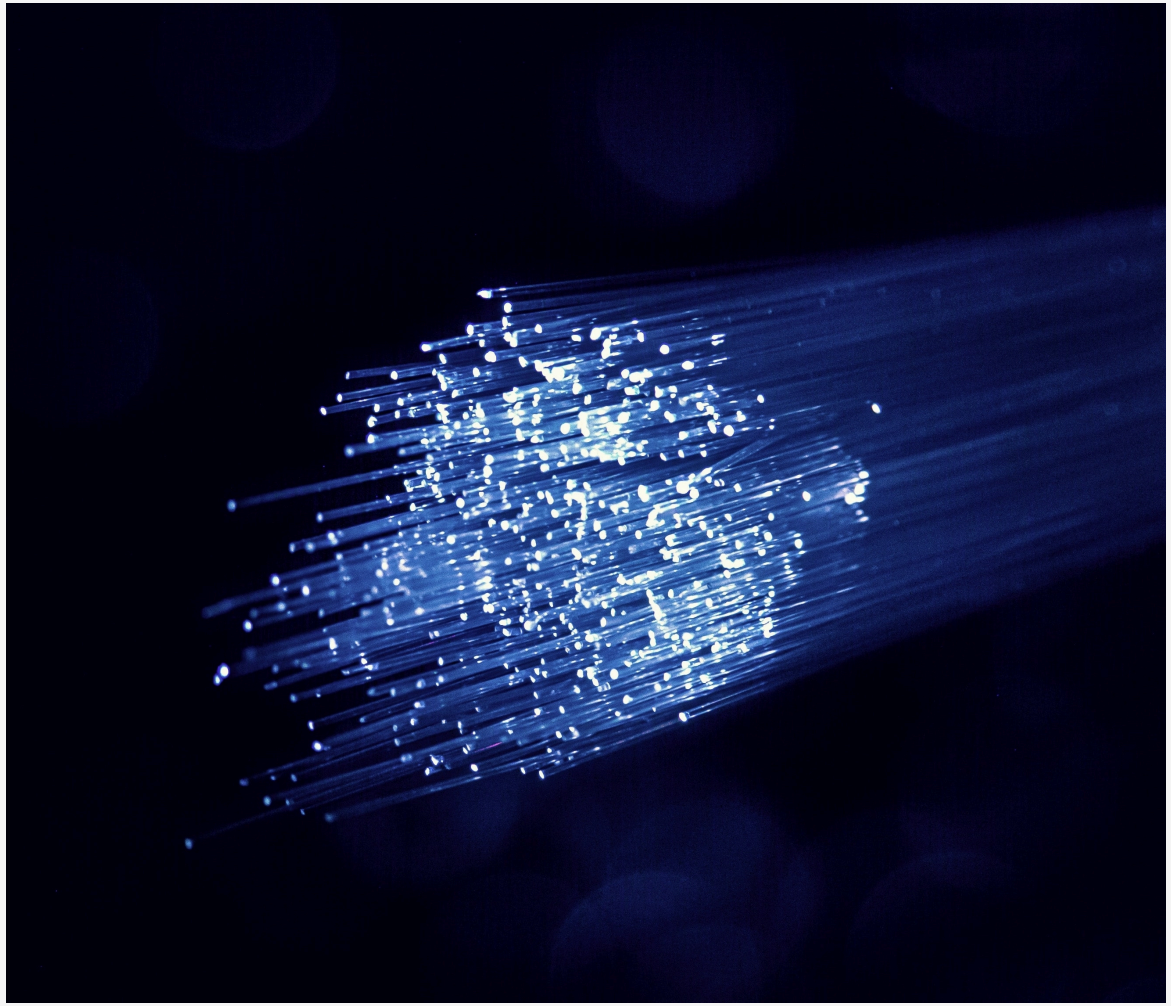
- Before P: background noise
- P and S: short arrivals over a wide band
- Surface waves: long, large, low frequency
- Dispersion: the frequency of the surface waves increases with time



Kermadec Islands earthquake (Mw 8.1, 4 March 2021) recorded at IU.CTAO.00.BHZ, windows of 100 s as in the lab, power relative to its median at each frequency

Cutting-edge seismology

- New sensors: fiber optics, nodes, smartphones
- Dense networks, and big data
- Open data and open software
- Machine learning and high-performance computing



Optical fibers, the sensors of distributed acoustic sensing

Software

Obspy: a Python toolbox for seismology.

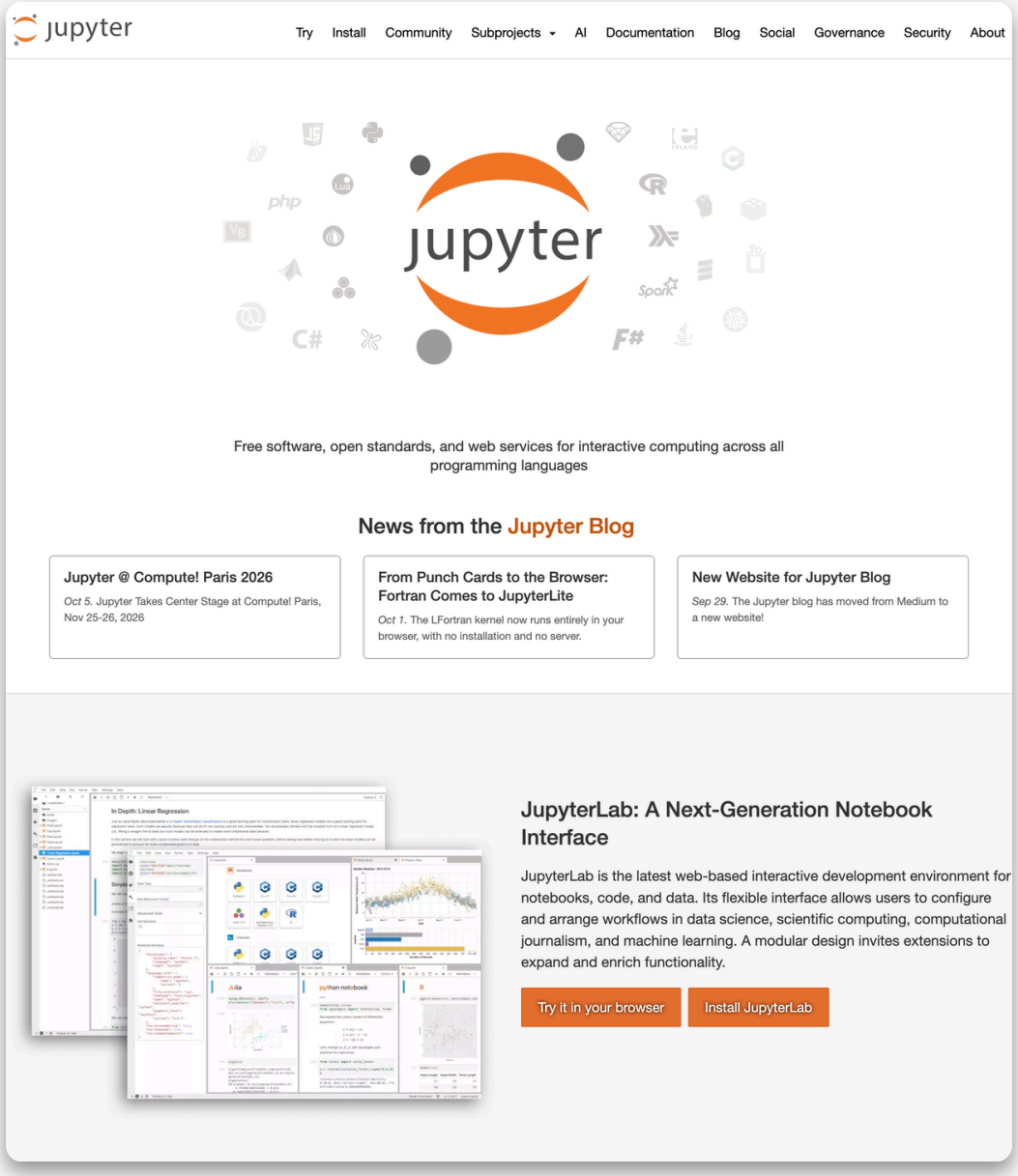
- Documentation and tutorials: docs.obspy.org
- Download data from datacenters
- Read most seismological formats
- Process and plot seismograms
- Compute travel times and ray paths
- Tutorials and examples



Beyreuther, M., Barsch, R., Krischer, L., Megies, T., Behr, Y., & Wassermann, J. (2010). ObsPy: A Python toolbox for seismology. *Seismological Research Letters, 81*(3), 530–533.

Scientific Python

- NumPy: arrays, linear algebra, Fourier transforms (`numpy.fft`)
- SciPy: signal processing (`scipy.signal`)
- Matplotlib: publication-quality plots (matplotlib.org)
- Jupyter: notebooks mixing code, figures and text. VS Code runs them too

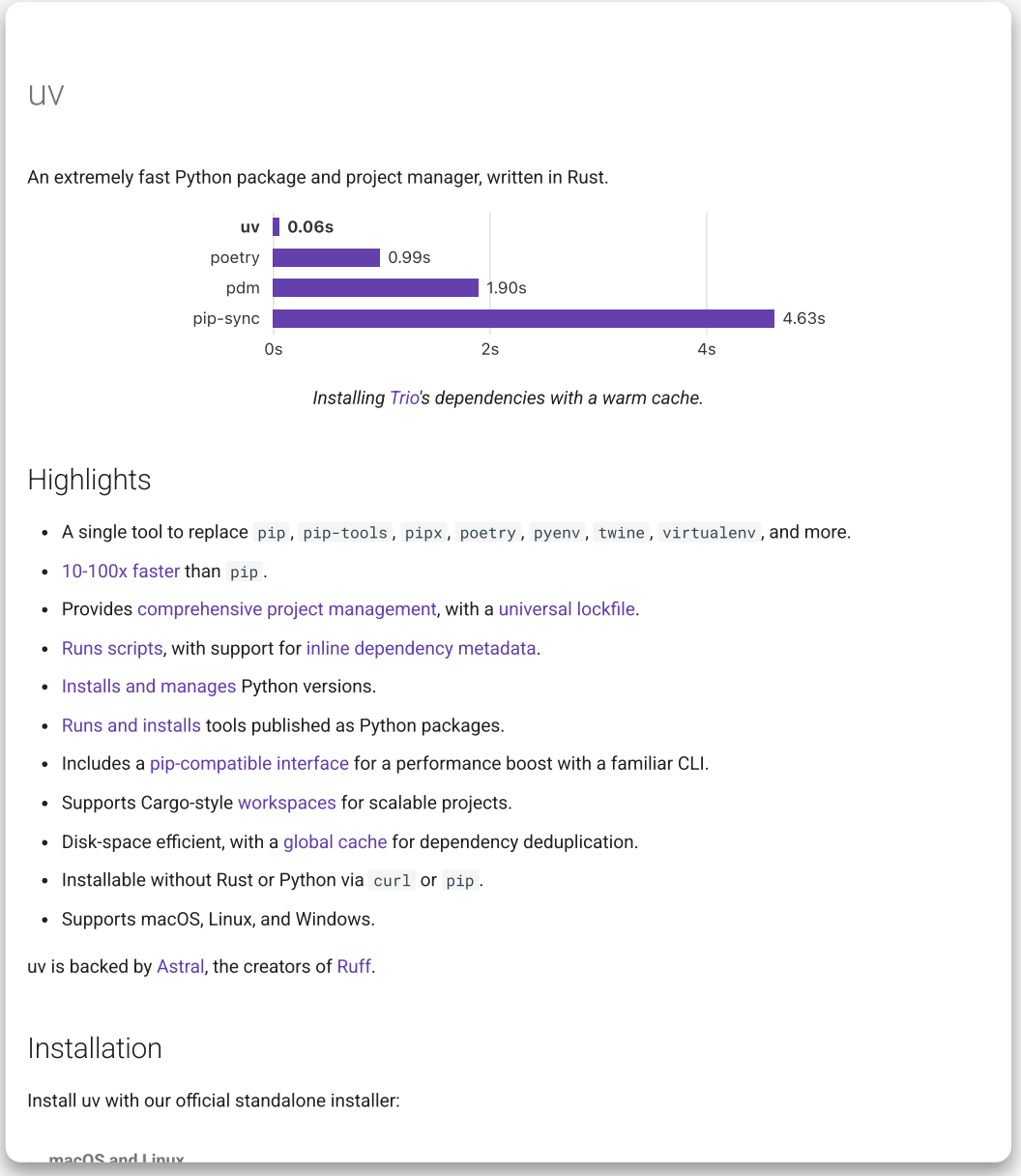


Harris et al. (2020), *Nature*, NumPy. Virtanen et al. (2020), *Nature Methods*, SciPy. Hunter (2007), *Computing in Science & Engineering*, Matplotlib. Kluyver et al. (2016), Jupyter Notebooks.

Managing packages

- Installs Python and packages, very fast
- One virtual environment per project
- Lock file for reproducible environments
- Runs scripts with `uv run`

Alternatives include Anaconda (`conda`), `pip` with `venv`, or Docker



uv

An extremely fast Python package and project manager, written in Rust.

Installing Trio's dependencies with a warm cache.

Highlights

- A single tool to replace `pip`, `pip-tools`, `pipx`, `poetry`, `pyenv`, `twine`, `virtualenv`, and more.
- 10-100x faster than `pip`.
- Provides comprehensive project management, with a universal lockfile.
- Runs scripts, with support for inline dependency metadata.
- Installs and manages Python versions.
- Runs and installs tools published as Python packages.
- Includes a pip-compatible interface for a performance boost with a familiar CLI.
- Supports Cargo-style workspaces for scalable projects.
- Disk-space efficient, with a global cache for dependency deduplication.
- Installable without Rust or Python via `curl` or `pip`.
- Supports macOS, Linux, and Windows.

uv is backed by Astral, the creators of Ruff.

Installation

Install uv with our official standalone installer:

macOS and Linux

Astral Software Inc. (2024). *uv: An extremely fast Python package and project manager* [Computer software]. https://github.com/astral-sh/uv

Computing at IPGP

S-CAPAD, the computing service of IPGP, runs the DANTE supercomputer, shared with the APC laboratory since 2021

- More than 7,000 CPU cores, and GPUs
- 256 to 512 GB of memory per node
- About 300 Tflops at peak, a laptop about 1
- About 1 PB of parallel storage
- Slurm scheduler and JupyterHub

Figures from IPGP and Université Paris Cité (2021–2026)

The Zen of Python

Beautiful is better than ugly.
Explicit is better than implicit.
Simple is better than complex.
Complex is better than complicated.
Flat is better than nested.
Sparse is better than dense.
Readability counts.
Special cases aren't special enough to break the rules.
Although practicality beats purity.
Errors should never pass silently.
Unless explicitly silenced.
In the face of ambiguity, refuse the temptation to guess.
There should be one-- and preferably only one --obvious way to do it.
Although that way may not be obvious at first unless you're Dutch.
Now is better than never.
Although never is often better than *right* now.
If the implementation is hard to explain, it's a bad idea.
If the implementation is easy to explain, it may be a good idea.
Namespaces are one honking great idea -- let's do more of those!

Type `import this` in Python. Read the PEP 8 guidelines before naming `f` your frequency and `t` your time

PEP 8 – Style Guide for Python Code

Source: https://github.com/python/peps/blob/main/peps/pep-0008.rst

Last modified: 2025-04-04 00:19:04 UTC

Author: Guido van Rossum <guido at python.org>, Barry Warsaw <barry at python.org>, Alyssa Coghlan <ncoghlan at gmail.com>
Status: Active
Type: Process
Created: 05-Jul-2001
Post-History: 05-Jul-2001, 01-Aug-2013

Introduction (#introduction)

This document gives coding conventions for the Python code comprising the standard library in the main Python distribution. Please see the companion informational PEP describing style guidelines for the C code in the C implementation of Python (../pep-0007/).

This document and PEP 257 (../pep-0257/) (Docstring Conventions) were adapted from Guido's original Python Style Guide essay, with some additions from Barry's style guide [2] (#id6).

This style guide evolves over time as additional conventions are identified and past conventions are rendered obsolete by changes in the language itself.

Many projects have their own coding style guidelines. In the event of any conflicts, such project-specific guides take precedence for that project.

A Foolish Consistency is the Hobgoblin of Little Minds (#a-foolish-consistency-is-the-hobgoblin-of-little-minds)

One of Guido's key insights is that code is read much more often than it is written. The guidelines provided here are intended to improve the readability of code and make it consistent across the wide spectrum of Python code. As PEP 20 (../pep-0020/) says, "Readability counts".

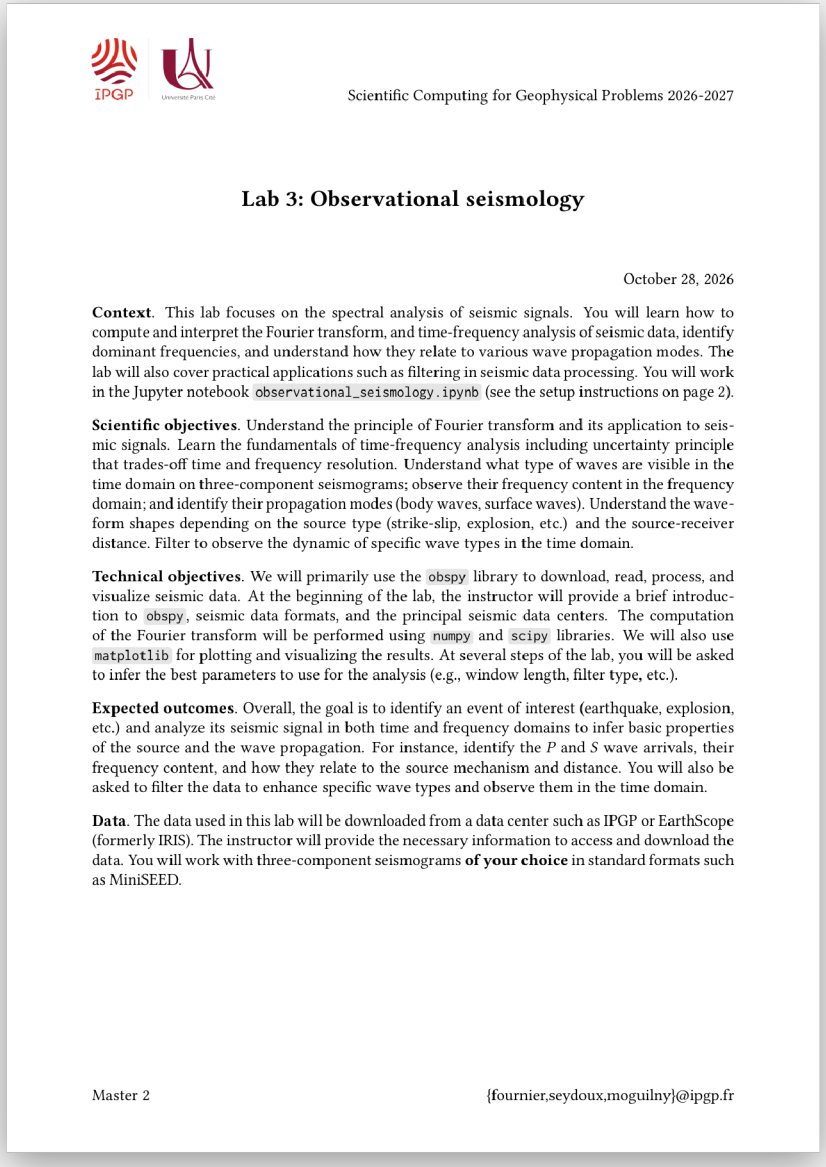
A style guide is about consistency. Consistency with this style guide is important. Consistency within a project is more important. Consistency within one module or function is the most important.

However, know when to be inconsistent – sometimes style guide recommendations just aren't applicable. When in doubt, use your best judgment. Look at other examples and decide what looks best. And don't hesitate to ask!

In particular: do not break backwards compatibility just to comply with this PEP!

Hands-on time

Read the instructions on the paper sheet I just distributed



IPGP Université Paris Cité

Scientific Computing for Geophysical Problems 2026-2027

Lab 3: Observational seismology

October 28, 2026

Context. This lab focuses on the spectral analysis of seismic signals. You will learn how to compute and interpret the Fourier transform, and time-frequency analysis of seismic data, identify dominant frequencies, and understand how they relate to various wave propagation modes. The lab will also cover practical applications such as filtering in seismic data processing. You will work in the Jupyter notebook `observational_seismology.ipynb` (see the setup instructions on page 2).

Scientific objectives. Understand the principle of Fourier transform and its application to seismic signals. Learn the fundamentals of time-frequency analysis including uncertainty principle that trades-off time and frequency resolution. Understand what type of waves are visible in the time domain on three-component seismograms; observe their frequency content in the frequency domain; and identify their propagation modes (body waves, surface waves). Understand the waveform shapes depending on the source type (strike-slip, explosion, etc.) and the source-receiver distance. Filter to observe the dynamic of specific wave types in the time domain.

Technical objectives. We will primarily use the `obspy` library to download, read, process, and visualize seismic data. At the beginning of the lab, the instructor will provide a brief introduction to `obspy`, seismic data formats, and the principal seismic data centers. The computation of the Fourier transform will be performed using `numpy` and `scipy` libraries. We will also use `matplotlib` for plotting and visualizing the results. At several steps of the lab, you will be asked to infer the best parameters to use for the analysis (e.g., window length, filter type, etc.).

Expected outcomes. Overall, the goal is to identify an event of interest (earthquake, explosion, etc.) and analyze its seismic signal in both time and frequency domains to infer basic properties of the source and the wave propagation. For instance, identify the *P* and *S* wave arrivals, their frequency content, and how they relate to the source mechanism and distance. You will also be asked to filter the data to enhance specific wave types and observe them in the time domain.

Data. The data used in this lab will be downloaded from a data center such as IPGP or EarthScope (formerly IRIS). The instructor will provide the necessary information to access and download the data. You will work with three-component seismograms **of your choice** in standard formats such as MiniSEED.

Master 2

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